

Bachelor of Science in Electrical and Electronic Engineering
EEE 400 (July 2025): Thesis

**Multi-Task Deep Learning for Atrial Fibrillation Detection and
Diagnostic Classification of 12-Lead ECG with Explainable Attribution**

Submitted by

Sadman Samin Rahul
2006055

Md. Shahriar Hasan
2006164

Supervised by

Dr. Hafiz Imtiaz
Professor



Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh

Jun 2026

CANDIDATES' DECLARATION

This is to certify that the work presented in this thesis is the outcome of the investigation and research carried out by us under the supervision of Dr.Hafiz Imtiaz. It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma or other qualifications.

Sadman Samin Rahul
2006055

Md. Shahriar Hasan
2006164

CERTIFICATION

This thesis, titled 'Multi-Task Deep Learning for Atrial Fibrillation Detection and Diagnostic Classification of 12-Lead ECG with Explainable Attribution', submitted by the group as mentioned below, has been accepted as satisfactory in partial fulfillment of the requirements for EEE 400: Project/Thesis, and as the requirements for the degree B.Sc. in Electrical and Electronic Engineering in May 2025.

Group Members:

Sadman Samin Rahul

Md. Shahriar Hasan

Supervisor:

Dr.. Hafiz Imtiaz

Professor

Department of Electrical and Electronic Engineering

Bangladesh University of Engineering and Technology

Table 1: Thesis Outcome Evidence and Engineering Complexity Justification (Course Outcomes).

CO	Course Outcome Statement	Mapped PO	Evidence from Thesis Work	Thesis Reference	Attained?
CO1	Identify, formulate, re-search literature, interpret data, and analyze problems using principles of mathematical, natural and engineering sciences.	PO(b)	The introduction frames automated AFIB detection as a problem of severe class imbalance and model opacity, motivated from cardiac electrophysiology and the limits of manual ECG reading. The research gap is established from a structured literature review, and the PTB-XL data and its label construction define the analysis basis.	Chapter 1 (1.1); Chapter 2 (2.5, Table 2.1); Chapter 3 (3.2.1)	☐
CO2	Investigate problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	PO(d)	A full experimental methodology is designed: stratified-fold training, validation-derived thresholding (Youden's J), internal test evaluation, and external cross-dataset validation on Chapman–Shaoxing. Results are interpreted through ROC/PR analysis and a deletion/insertion faithfulness study, and synthesised into validated conclusions about detection and interpretability.	Chapter 3 (3.4); Chapter 4 (4.4, 4.5, Tables 4.2–4.5); Chapter 4 (4.7)	☐

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CO	Course Outcome Statement	Mapped PO	Evidence from Thesis Work	Thesis Reference	Attained?
CO3	Apply knowledge of mathematics, science, and engineering to solve research problems.	PO(a)	Engineering knowledge is applied throughout: a 1-D CNN–BiLSTM architecture, signal preprocessing (0.5 Hz high-pass filtering, per-lead z-score normalisation), an imbalance-handling sampler and weighted loss, and attribution mathematics (Integrated Gradients, Grad-CAM, occlusion, feature ablation). Design choices such as kernel/stride configuration and the recall-oriented operating point are justified.	Chapter 3 (3.2.2–3.2.4, Table 3.1); Chapter 3 (3.4.2)	☐
CO4	Demonstrate adherence to professional ethics and ethical principles of research.	PO(h)	The work uses only public, de-identified datasets in accordance with their terms, makes no attempt at re-identification, and cites all sources, prior work, and libraries. Less flattering results (e.g. the lower precision from the recall-oriented threshold and the precision drop on Chapman) are reported honestly rather than omitted.	Chapter 5 (5.3); Chapter 5 (5.1.3); Bibliography	☐

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CO	Course Outcome Statement	Mapped PO	Evidence from Thesis Work	Thesis Reference	Attained?
CO5	Communicate effectively with oral presentation and detailed thesis report.	PO(j)	Findings are communicated through a structured thesis report with clearly labelled figures and tables, system and architecture diagrams, attribution heatmaps, and template-based natural-language explanations that render technical attribution into plain language for a non-specialist reader.	Full report; Figures 3.1, 4.1–4.16; Tables 4.1–4.5; Presentation slides	☐
CO6	Research contemporary aspects related to the problem and topic through detailed literature review and understand.	PO(i)	A detailed review covers contemporary work on deep learning for ECG, the PTB-XL benchmark, imbalanced-learning methods, and explainable AI, situating the contribution against recent studies. Current research trends (transformers, trustworthy XAI, LoRA) are discussed, and concrete future-work directions are identified.	Chapter 2 (2.1–2.5); Chapter 5 (5.4); Bibliography (38 refs)	☐

Table 2: Complex Engineering Problem and Activity Attributes addressed by this thesis.

Code	Meaning	Thesis-Specific Justification
Knowledge Profile Attributes		
K1	Knowledge of natural sciences applicable to the engineering discipline.	<i>The work rests on cardiac electrophysiology: atrial fibrillation manifests in the 12-lead ECG through irregular R–R intervals, absent P-waves, and fibrillatory activity best seen in lead VI. These physiological signatures underpin the labelling, the model design, and the clinical interpretation of the attribution results.</i>
K2	Conceptually based mathematics, numerical analysis, statistics, and formal aspects of computer and information science supporting analysis and modelling.	<i>Heavy use of deep-learning mathematics (1-D convolution, bidirectional LSTM recurrence, gradient-based optimisation with AdamW), signal processing (0.5 Hz high-pass filtering, per-lead z-score normalisation, resampling), probability and statistics (weighted binary cross-entropy, ROC-AUC, F1, precision/recall, Youden’s J), HRV measures (RMSSD, pNN50, RR coefficient of variation, LF/HF ratio), and attribution mathematics (Integrated Gradients path integrals, Grad-CAM, deletion/insertion AUC).</i>
K3	Engineering fundamentals required for the engineering discipline.	<i>Core EEE fundamentals: biomedical signal acquisition and conditioning, sampling theory (500 Hz, 10-second/5000-sample windows), digital filtering, time-series analysis, and the design and training of neural networks as signal classifiers — the foundational basis of the detector.</i>
K4	Specialist knowledge that provides theoretical frameworks and bodies of knowledge for accepted practice areas in the engineering discipline.	<i>Specialist knowledge in deep learning for ECG interpretation: the CNN–BiLSTM hybrid that captures local beat morphology and longer rhythm context, imbalance-learning theory (weighted sampling and cost-sensitive loss), and explainable AI (gradient- and perturbation-based attribution, faithfulness evaluation), together with the PTB-XL benchmarking framework.</i>

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Code	Meaning	Thesis-Specific Justification
K5	Engineering design knowledge that supports the design of complex engineering solutions.	<i>End-to-end design of a multi-component system: a 1-D CNN encoder feeding a two-layer BiLSTM with a compact detection head, a stratified-fold data pipeline, a two-front imbalance-handling scheme (weighted sampler with inverse-frequency buckets, $1.5\times$ boost, and 5.0 cap, plus a class-weighted loss), and a template-based natural-language explanation pipeline — all designed and integrated by the authors.</i>
K6	Knowledge of engineering practice, including technology, tools, resources, safety, standards, and practical constraints.	<i>Practical use of professional tools and resources: Python and PyTorch for model development, the Cap-tum library for attribution, GPU training with mixed-precision arithmetic and gradient clipping, public clinical databases (PTB-XL and Chapman–Shaoxing), and standardised evaluation protocols. Practical constraints such as class imbalance, compute budget, and threshold selection were handled explicitly.</i>
K7	Comprehension of the role of engineering in society, including ethics, sustainability, economy, environment, health, safety, and public welfare.	<i>The thesis explicitly weighs health and safety (recall-oriented screening where a missed arrhythmia is the costlier error), societal fairness (risk of underperformance on under-represented populations), ethics and patient-data handling (use of public, de-identified datasets), and sustainability (a deliberately compact model and template-based explanation to keep the computational footprint low).</i>
K8	Engagement with research literature in the discipline.	<i>The work is grounded in a substantial body of recent literature — 38 references including the PTB-XL dataset and benchmark papers, foundational ECG deep-learning studies, the SMOTE and focal-loss imbalance literature, and the Integrated Gradients, Grad-CAM, occlusion, LoRA, and xECGArch explainability papers — used to position the contribution and identify the research gap.</i>

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Code	Meaning	Thesis-Specific Justification
Complex Engineering Problem Attributes		
P1	Depth of knowledge required: the problem cannot be solved without applying principles from mathematics, natural sciences, engineering fundamentals, and specialist engineering knowledge.	<i>Solving the problem demanded deep, integrated knowledge rather than routine application: combining cardiac electrophysiology, deep sequence modelling, imbalanced-learning theory, and attribution mathematics. Naive application fails — a classifier that ignores the imbalance reaches ~93% accuracy while catching no AFIB cases, so principled design choices were essential.</i>
P2	Range of conflicting requirements: the problem involves wide-ranging or conflicting technical, performance, cost, safety, environmental, social, or practical requirements.	<i>The central conflict is accuracy versus interpretability — the most accurate models are usually the least transparent, yet the thesis pursued both at once. A second trade-off is precision versus recall: the screening role deliberately favours recall (93%) at the cost of lower precision (0.744), accepting more false alarms to avoid the costlier missed arrhythmia.</i>
P3	Depth of analysis required: the problem needs analysis, modelling, simulation, experiment, interpretation, or judgment beyond a standard textbook solution.	<i>The problem required extensive analysis beyond a textbook solution: training-dynamics analysis, threshold selection via Youden's J, internal test evaluation, external cross-dataset validation on Chapman-Shaoxing, multi-method attribution analysis, global lead-importance aggregation over 20 cases, and a quantitative deletion/insertion faithfulness study over 30 recordings.</i>
P4	Familiarity of issues: the problem may include infrequently encountered, partially defined, emerging, or research-oriented issues.	<i>The work engages emerging, research-oriented issues: trustworthy and explainable AI for medicine is an active and unsettled research area, and combining imbalance-aware detection with multi-method cross-checked attribution and natural-language explanation in one system is a gap not previously closed for atrial fibrillation.</i>
P5	Extent of applicable codes: the problem may require codes, standards, engineering practices, or literature that are not fully sufficient by themselves.	<i>No single standard or recipe fully governs the task. The PTB-XL recommended stratified folds and SNOMED-CT rhythm annotations (e.g. AFIB code 164889003) provided structure, but design rules for imbalance handling, threshold setting, and attribution faithfulness had to be assembled from the literature rather than applied off the shelf.</i>

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Code	Meaning	Thesis-Specific Justification
P6	Extent of stakeholder involvement and conflicting requirements: the problem may involve users, supervisors, clients, society, industry, or other stakeholders with different needs.	<i>Multiple stakeholders shape the requirements: clinicians (who need inspectable, trustworthy decisions and retain override), patients (for whom a missed arrhythmia is dangerous), and healthcare systems (which need scalable triage). Their differing needs motivated the recall-oriented operating point, the explanation component, and the framing of the tool as decision support under expert oversight.</i>
P7	Interdependence: the problem has multiple connected components, variables, sub-systems, or constraints that influence one another.	<i>The system has tightly coupled components: the CNN encoder feeds the BiLSTM, whose pooled features drive the detection head; the weighted sampler and the class-weighted loss jointly control imbalance; the validation-derived threshold links the validation, internal-test, and external-test stages; and the attribution outputs feed directly into the natural-language explanation generator. Changing one element propagates through the others.</i>

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Code	Meaning	Thesis-Specific Justification
Complex Engineering Activity Attributes		
A1	Range of resources: the activity uses diverse resources such as people, equipment, software, data, literature, laboratory facilities, or engineering tools.	<i>The activity drew on diverse resources: two public 12-lead ECG databases (PTB-XL, 21,837 records; Chapman–Shaoxing, 10,247 records), software libraries (PyTorch, Captum), GPU computing hardware, a frozen pre-trained language model for explanation, and an extensive body of research literature.</i>
A2	Level of interaction: the activity requires interaction among tasks, tools, systems, people, or disciplines.	<i>The work required interaction across tasks and disciplines: coordinating modelling with validation (internal fold and external dataset), coupling the detection model with the attribution and explanation stages, cross-checking four independent attribution methods against one another, and combining clinical-physiological reasoning with machine-learning practice.</i>
A3	Innovation: the activity involves creative use of engineering knowledge, adaptation of methods, new design choices, or non-routine problem solving.	<i>The original contributions include a compact imbalance-aware CNN–BiLSTM AFIB detector, a multi-method attribution analysis whose cross-method agreement is used as evidence of genuine feature reliance, a quantitative faithfulness evaluation, and a faithful-by-construction template generator that turns attribution into plain-language explanation — with the data structured for future LoRA-based learned generation.</i>
A4	Consequences for society and environment: the activity may have effects on health, safety, sustainability, environment, economy, or society.	<i>The outcome could extend cardiac screening to settings lacking specialist cardiologists and reduce redundant testing, while the compact design keeps energy cost low. The thesis also names the risks — potential underperformance on under-represented groups and the danger of acting on false positives — and limits the system to decision support under expert oversight.</i>

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Code	Meaning	Thesis-Specific Justification
A5	Familiarity: the activity may require work in unfamiliar, uncertain, or changing contexts.	<i>The work was carried out under genuine uncertainty: generalisation to an unseen population and acquisition setup was tested rather than assumed (and showed a precision drop and a shifted lead ranking on Chapman), attribution methods can disagree at the margins, and the SNOMED-CT label mapping for the external set had to be derived carefully. These unfamiliar, changing conditions were handled and reported honestly.</i>

I/We, the undersigned, pledge that I/we filled out the tables with consultation and feedback of my/our supervisor.

Sadman Samin Rahul

ID: 2006055

Shahriar Hasan

ID: 2006164

The relevant tables are verified based on the thesis report, presentation, demonstration, viva voce, and assessment evidence. Any suggested modifications are reflected in the table.

Dr. Hafiz Imtiaz

Professor

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Dhaka

Sadman Samin Rahul

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Md. Shahriar Hasan

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ABSTRACT

Cardiovascular disease remains one of the leading causes of death worldwide. The electrocardiogram (ECG) is still the first tool clinicians reach for when screening the heart. Reading a 12-lead trace correctly, however, takes training and time. And even experienced readers disagree on subtle cases. Atrial fibrillation (AFIB), one of the most common and clinically important arrhythmias, is a natural target for automated screening. But two problems make automated reading harder than it first appears: in any realistic recording set the arrhythmia is heavily outnumbered by normal traces, and the most accurate models tend to behave like black boxes, which clinicians are reluctant to trust. This thesis sets out to tackle both at once. To build a model that detects atrial fibrillation reliably on imbalanced data and that can say, in plain terms, why it reached a given decision.

We work with the PTB-XL database, a publicly available collection of 21,837 clinical 12-lead recordings sampled at 500 Hz. The detector is a single network combining a one-dimensional convolutional encoder with a bidirectional LSTM, capturing both local waveform morphology and the longer rhythm context that distinguishes a fibrillating heart. Because AFIB accounts for under 7% of the recordings, the class imbalance is addressed directly through a weighted sampler that rebalances each training batch together with a weighting in the loss, so that the rare positive cases are not drowned out by the abundant normal ones. To open up the model, we apply four gradient- and perturbation-based attribution methods—Integrated Gradients, Grad-CAM, occlusion, and feature ablation—to identify which leads and time windows drove each prediction, and we convert these attributions into short, readable explanations using a template-based text generator.

The detector performs strongly. On the held-out test set it reaches a ROC-AUC of 0.979 and recalls about 93% of true AFIB cases at the default decision threshold, a useful result for a class making up under 7% of the data. The lower precision that accompanies this high recall is an acceptable trade for a screening role, where a missed arrhythmia is the costlier error. The explainability analysis is equally encouraging. For a correctly detected case the four attribution methods agree closely, placing the strongest evidence on clinically sensible leads—V1, aVR, and Lead II prominent among them—and on the regions around individual heartbeats rather than on arbitrary artefacts. Averaged over many AFIB-positive recordings, the same leads rise to the top, indicating a consistent and repeatable preference rather than a one-off. That four methods built on different principles converge on the same evidence suggests the network is keying on real features, and the generated explanations summarise that evidence faithfully in plain language. A second head shares the same encoder and is trained in a multi-label fashion to classify each recording into one of 5 diagnostic super-

classes (NORM, MI, CD, HYP, STTC); thus, a single network is used for detection AND diagnostic classification. This auxillary head achieves a held-out macro-AUC of 0.908 on the held-out test fold, and the rare hypertrophy class is the most challenging to capture.

Keywords: electrocardiography, PTB-XL, atrial fibrillation, deep learning, CNN-BiLSTM, class imbalance, explainable AI, attribution methods.

MAIN BODY

CHAPTER 1:

INTRODUCTION

The heart announces its faults in a language of small electrical signals. For more than a century the electrocardiogram (ECG) has been the standard way of listening in. A twelve-lead recording takes seconds to capture, costs very little, and carries a remarkable amount of information about rhythm, conduction, and the state of the heart muscle. Still the gap between recording an ECG and reading it correctly is wide. Interpretation still leans on trained clinicians. Expert time is limited, and inter-reader disagreement is frequent, even on cases that appear straightforward in hindsight [3]. As wearable monitors and portable devices push the number of recordings far beyond what any cardiology department can review by hand [30], the case for reliable automated interpretation has only grown stronger.

Among the conditions worth detecting automatically, atrial fibrillation (AFIB) stands out. It is one of the most prevalent persistent arrhythmias [2]. It significantly increases the risk of stroke and heart failure [1]. And it is often intermittent and easy to overlook on a brief recording. This makes it a natural target for a screening tool that can sort through volumes of ECGs and flag the ones that need more investigation. Deep learning has made real progress on exactly this kind of task. With convolutional and recurrent models now matching or surpassing clinicians on several narrow problems. Two stubborn obstacles remain though, and this thesis is built around them.

The first is imbalance. In any realistic collection of ECGs the arrhythmia of interest is heavily outnumbered by normal recordings. In the dataset used here, atrial fibrillation appears in fewer than one recording in every fourteen. so a model trained carelessly can post a deceptively high accuracy by leaning toward the majority while missing the very cases that matter. The second is opacity. The most accurate models tend to be the least transparent, and a black-box verdict, however correct, is a hard thing to act on in a clinical setting. Where someone has to take responsibility for the decision. A tool that cannot show its reasoning is a tool that will sit unused.

1.1 Problem Statement and Motivation

This work asks whether a single, compact model can detect atrial fibrillation reliably on strongly imbalanced data. While simultaneously being able to point back to the part of the signal that justified its answer. The two requirements are usually conflicting. The techniques that push accuracy highest tend to make a model harder to interpret. The contribution of this thesis is to pursue both together rather than trading one for the other. The motivation is

practical as much as technical: a detector that is dependable on the rare positive class and can self-explain is far closer to something a clinician might actually trust at the bedside, than one that is merely accurate.

Concretely, the thesis sets out to do three things. To build a detector, combining a convolutional encoder with a bidirectional recurrent layer, that picks atrial fibrillation out of a twelve-lead ECG. To confront the class imbalance head-on through a weighted sampling scheme so that the rare positive cases are not lost. And to make the model’s decisions inspectable by applying several attribution methods to identify the leads and time windows behind each prediction. And then rendering those findings into short, readable explanations. The scope is deliberately bounded to atrial fibrillation and to a single large public dataset. A choice revisited when the results and their limits are discussed.

1.2 Contributions

The specific contributions of this thesis can be summarised as follows:

- **A compact, imbalance-aware atrial fibrillation detector.** A single CNN–BiLSTM model is trained to detect atrial fibrillation from the raw twelve-lead signal. Using a weighted sampling scheme to confront a strong class imbalance in which the positive class makes up under 7% of recordings. On the held-out test fold it reaches a ROC-AUC of 0.979 and recalls roughly 93% of true cases.
- **A multi-method attribution analysis of the detector’s decisions.** Four attribution methods- Integrated Gradients, Grad-CAM, occlusion, and feature ablation are applied to the trained model. And their agreement is used as evidence that the highlighted leads and time windows reflect genuine signal features rather than artefacts of any one technique.
- **Evidence that the detector relies on clinically sensible features.** Both on individual cases and averaged across a population of atrial-fibrillation-positive recordings, the model’s attribution concentrates consistently on rhythm-relevant leads- V1, aVR, and Lead II foremost - and on the regions around individual heartbeats.
- **A lightweight pipeline that turns attribution into plain-language explanation.** The condensed attribution findings are rendered into short, readable explanations by a template-based generator that is faithful to the evidence by construction, with each explanation built as an instruction-style record so the data is ready for learned generation – for instance via parameter-efficient fine-tuning – as a future extension.

The remainder of this thesis is organised as follows. Chapter 2 reviews the relevant work on deep learning for ECG interpretation, the PTB-XL benchmark, learning under class

imbalance, and explainable AI. Chapter 3 sets out the dataset, model, training procedure, and explanation pipeline. Chapter 4 reports the detection and explainability results. And Chapter 5 weighs the broader societal, ethical, and sustainability dimensions of the work and the directions it points toward.

CHAPTER 2:

LITERATURE REVIEW

The idea of teaching a machine to read an electrocardiogram is not new. But the last decade has changed what that ambition looks like in practice. This chapter traces that shift across the four strands that this thesis draws on. The move from hand-crafted features to deep learning for ECG interpretation, the datasets and benchmarks that made fair comparison possible, the long-standing problem of learning from imbalanced clinical data, and the more recent push to make these models explain themselves. Reviewing each in turn makes clear both where the field has settled and where the gaps this work tries to fill still sit.

2.1 Deep Learning for ECG Interpretation

For a long time, automated ECG analysis rested on a pipeline of careful preprocessing, beat segmentation, and engineered features fed to a conventional classifier [16]. The limitation was always the same. Performance was capped by how good the hand-designed features were. And those features rarely transferred cleanly from one condition or dataset to another. The turning point came when end-to-end deep networks began to learn their own representations directly from the raw signal. Hannun et al. [5] were among the most influential here, training a deep convolutional network on more than ninety thousand single-lead recordings and reporting rhythm-classification performance on par with a panel of board-certified cardiologists. The same conclusion was later reached on full twelve-lead recordings by Ribeiro et al. [23]. on a dataset of more than two million exams. That result did a great deal to convince the community that learned features could rival, and in places surpass, decades of expert feature engineering.

What followed was a steady refinement of architecture rather than a single leap [17]. Convolutional networks proved good at capturing the local morphology of a heartbeat - the shape of the QRS complex, the deflection of an ST segment. But a heartbeat in isolation says little about rhythm, which only reveals itself over a longer stretch of signal. This is especially true for atrial fibrillation. Whose hallmark is the irregularity of successive beats rather than the shape of any one of them. The observation motivated hybrid designs that pair a convolutional front end with a recurrent back end. So that local shape and longer-range temporal context are both modelled. Oh et al. [15] demonstrated this early on. Combining convolutional and LSTM layers to classify arrhythmias from variable-length ECG segments without hand-crafted feature extraction. Several groups have since reported that such CNN-LSTM combinations outperform either component on its own for arrhythmia tasks [8]. The

bidirectional variant, which reads the sequence in both directions, has been used to good effect for ECG classification [9]. More recently, attention-based and transformer architectures [36] have been applied to ECG analysis [34] [24]. Though they are typically more data-hungry than the convolutional–recurrent designs adopted here. The review here is partly an account of why the CNN-LSTM pairing has become a sensible default.

2.2 Datasets and Benchmarking on PTB-XL

Progress in any data-driven field is only as trustworthy as the data and the evaluation behind it. And for a long while ECG research suffered from a lack of both. Many early studies leaned on small or single-lead collections [18]. Which made it hard to know whether a reported number reflected a genuinely effective technique or simply a method that had memorised a limited dataset. The release of PTB-XL by Wagner et al. [6] changed this considerably. With 21,837 clinical twelve-lead recordings from close to nineteen thousand patients, annotated by cardiologists and carrying a rich set of diagnostic statements. It provided the community with a large, clinically grounded, and openly available reference point. And because those statements include rhythm labels such as atrial fibrillation, it supports binary arrhythmia-detection tasks just as easily as broader classification ones.

Crucially, PTB-XL arrived with recommended train-test splits. Which matters more than it might first appear. Without a fixed protocol, two papers reporting on the ‘same’ dataset may not be comparable at all. And patient overlap between training and test sets can subtly inflate results. Strodtz et al. [7] built on this by establishing a set of benchmark tasks and evaluating a range of modern architectures on them. Giving subsequent work a clear baseline to measure against. Their analysis also surfaced a practical lesson that this thesis takes seriously. That headline metrics can mask poor performance on the sparsely represented labels, Which are often precisely the ones of clinical interest. The dataset and these benchmarks together frame the experimental setting used in this work.

2.3 Learning from Imbalanced Clinical Data

The class-imbalance problem is not limited to cardiology but is of importance in this field. In any realistic The healthy recordings far outnumber the abnormal ones when collecting ECGs. And a specific A small proportion of the data may be due to arrhythmia, e.g. atrial fibrillation [20]. Under Here 7Overall good accuracy and some failure on the positive class. Simply because The loss function is easier to minimize if it leans towards the majority. For a screening tool this is the worst possible failure mode, since the rare cases are the main objective.

There are essentially two types of solutions [19]. Data-level methods balance what the model sees, by resampling, synthetic minority oversampling like SMOTE [13] or weighted.

A sampling that is biased in favor of the rare class. Rather than changing how mistakes are made, algorithm-level methods change the way mistakes are counted. Incorporating higher penalties for errors on the minority class, via class-weighted or Formulations based on focal loss [14]. In reality the two are frequently used together. This thesis leans chiefly on the data-level approach. To boost the rare positive class up to a level of A significant percentage of every cohort of training. The review located here contrasts that choice with the It is not meant to be the only choice but rather referred to in the broader literature.

2.4 Explainable AI

A good accuracy is not sufficient for clinical use. A model is, at its core, a representation of something — an attempt to capture the structure of a phenomenon in a form we can reason about. Correct answer without being able to show why is hard to believe, harder to debug and awkward to use. Respond to a question of judgement. These worries become more intense in a safety-critical environment. such as medicine [21]. This has led to continued interest in EAI. And in particular in attribution techniques which inquire about which parts of the input were most crucial for a particular prediction.

Several such methods are repeated in this work. And they fall into two families: Gradient-based methods and Grad-CAM. Gradient-based Integrated Gradients (Sundararajan et al. [10]) are some of the methods used. This approach uses a path from a to b to accumulate gradients over the input features, and assign a prediction to them. A formulation with attractive theoretical guarantees: baseline to the actual input. And Grad-CAM, The proposal by Selvaraju et al. [11] was suggested. This approach employs the flow gradients into a The convolutional layer was followed by a coarse map of the regions that contributed to the decision, which was generated using the convolutional layer. Other approaches, such as perturbation based methods like occlusion and feature ablation [22] follow a complementary approach. Evaluating the effect of masking or omitting portions of the input on the prediction. A number of these techniques have been originally popularized on images. However, they are naturally adaptable to one-dimensional where the question is which leads and which time windows were relevant. Encouragingly, in the case of Strodtz et al. [7] who used attribution for their PTB-XL models, The highlighted areas tended to correspond with clinically relevant areas of the trace. Suggesting that the Networks were using appropriate evidence.

Interpretability has since become more than an afterthought, it's a design goal. A One such clear example is the use of xECGArch proposed by Goettling et al. [4]. It's an architecture. Designed specifically for atrial fibrillation detection, a combination of short- and long-term. convolutional streams with explainability methods. And that systematically compares thirteen Point out to students that they can use perturbation analysis to find out which of the attribution techniques is the most effective. trustworthy explanations. They found

that different ways of attributing vary significantly in faithfulness is the direct motivation of cross-checking multiple methods, not trusting any one in particular. This thesis employs attribution in a similar way but takes it one step further. xECGArch is used to compare attribution methods and to display the areas of the trace where the attribution is significant. Whereas this work also supports the evidence with several methods and provides the result. As a comprehensible natural language explanation.

The last one is about the efficient generation of those explanations. Producing fluent text Fine-tuning from a large language model typically involves adjusting all of its parameters, which can be expensive. updated. Low-Rank Adaptation (LoRA) [12] avoids this by Freezing the pre-trained weights and training a small pair of low-rank matrices injected into. each layer. The number of the trainable parameters is reduced significantly while maintaining most of the The advantage of fully tuning. It is very efficient and well suited to the light explanation. component of this thesis. When the purpose is to state findings of attribution in simple terms Instead of creating a general-purpose model from the ground up.

2.5 Summary and Research Gap

Table 2.1: Comparison of representative ECG deep-learning studies against this work.

Study	12-lead	CNN-LSTM	Imbal. aware	xAI attr.	Multi-method	NL expl.
Hannun et al. [5]	–	–	–	–	–	–
Oh et al. [15]	–	✓	–	–	–	–
Cheng et al. [9]	–	✓	–	–	–	–
Strodthoff et al. [7]	✓	–	✓	✓	–	–
Goettling et al. [4]	–	–	–	✓	✓	–
This work	✓	✓	✓	✓	✓	✓

Abbr.: Imbal. = imbalance-handling; xAI attr. = attribution method;

Multi-method = cross-checks several attribution methods; NL expl. = natural-language explanation.

Read together, these strands point to a consistent picture. Deep learning, and CNN-LSTM hybrids in particular, can interpret ECGs to a high standard. PTB-XL and its benchmarks provide the data and the yardstick to do so credibly. Imbalance-aware training is essential if rare but important conditions are not to be lost. And attribution methods can expose a model’s reasoning. What is less common is work that brings all of these together in one system- an imbalance-aware detector that identifies a rare arrhythmia reliably and that not only highlights the evidence behind its predictions but corroborates that evidence across several attribution methods and renders it as human-readable explanation. Closing that gap,

for the specific case of atrial fibrillation, is the aim of this thesis. And the methodology in Chapter 3 describes how the pieces are assembled.

CHAPTER 3:

METHODOLOGY

This chapter sets out how the work was actually carried out, from the raw recordings through to the trained detector and its explanations. It begins with the theoretical ideas the design rests on, then describes the datasets and how the label was derived, the network itself, the training procedure, and finally the tools used to make the model’s decisions interpretable. The intent throughout is to give enough detail that the work could be reproduced, while flagging the choices that mattered most to the outcome.

3.1 Theoretical Framework

The initial hypothesis of this thesis is that there is a signature in the atrial fibrillation in the twelve-lead ECG that a model can be trained to recognise, but that recognising it well calls for simultaneous reading of the signal on two scales. The shape of a particular beat is part of the information, while the defining feature of fibrillation – an irregular, disorganised rhythm – only becomes apparent after a few beats. A model that only encompasses one scale will fail to capture what if the other holds, then a hybrid architecture is used here, a convolutional stage to read local waveform shape, followed by a recurrent stage to read the longer rhythm context.

The whole design is determined by a second commitment. A detector which yields a verdict with no, however accurate this is, it is of limited use in the clinical setting, as it is built on an account of its basis. interpretability is not considered as a diagnostic afterthought. The model is therefore designed to be probed: After being trained, decisions made by the model are explored with attribution. Techniques are used to identify parts of the signal that it used and these results are presented as plain language.

3.1.1 Signal Representation

A twelve-lead ECG is a multi-channel time series, and the two properties that matter for detecting atrial fibrillation live at different scales. The shape of an individual beat – the height of an R-wave, the form of the P-wave or its absence – is a local feature spanning tens of milliseconds. Rhythm, on the other hand, only becomes visible across several beats, over seconds of signal, and it is the irregularity of that rhythm that most distinguishes a fibrillating heart. A convolutional layer is well suited to the first scale, since its kernels learn to respond to local waveform shapes regardless of where in the trace they occur. A recurrent layer handles the second, carrying information forward – and, in the bidirectional case, backward

– across the sequence so that longer-range dependencies are not lost. The architecture used here deliberately stacks the two so that the convolutional front end distils each lead into a compact sequence of features, which the recurrent layer then reads as a temporal story.

3.2 Design

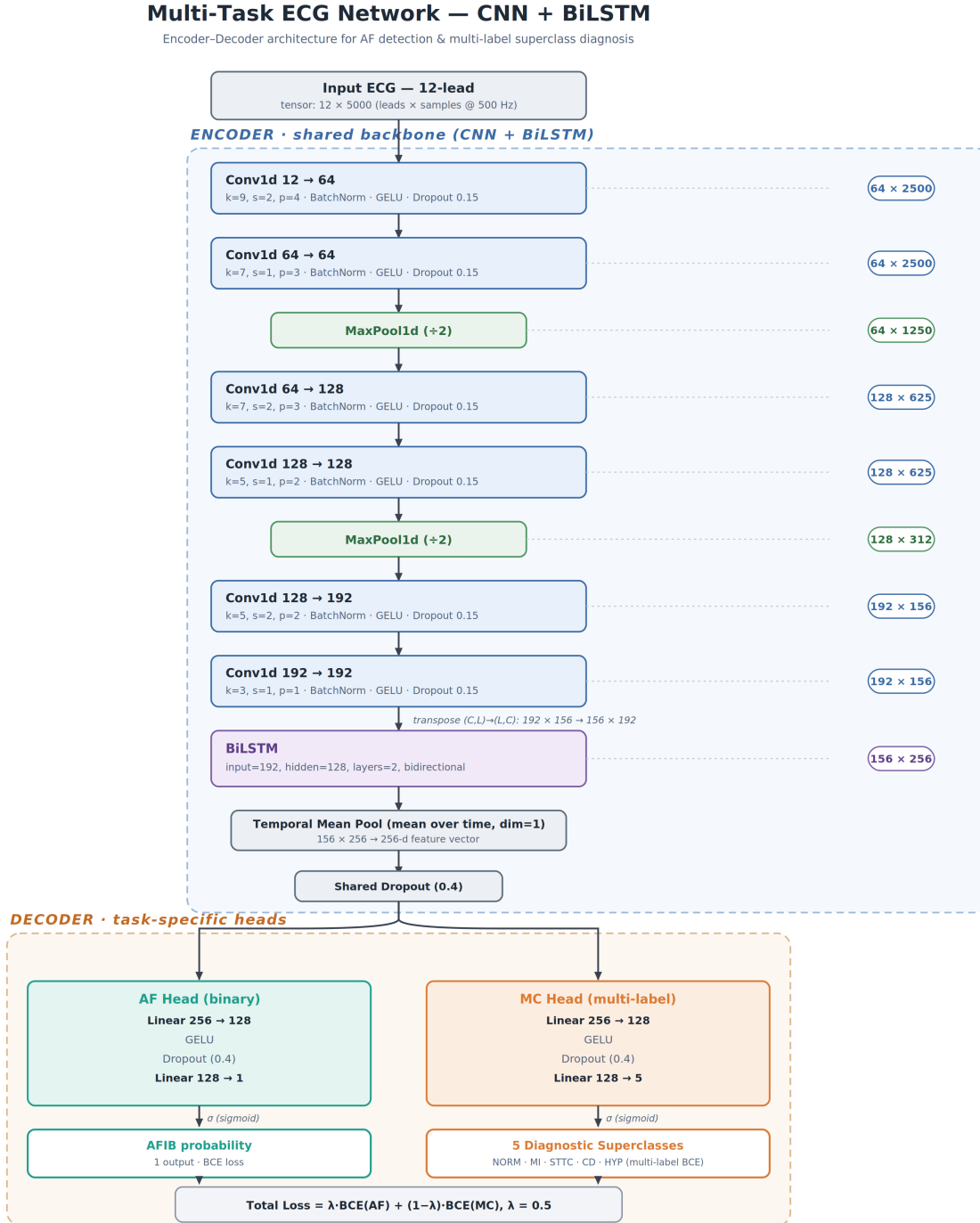


Figure 3.1: System overview of the proposed AFIB detection and explanation pipeline.

3.2.1 Dataset and Label Construction

The primary dataset is PTB-XL (version 1.0.1), a public collection of 21,837 clinical twelve-lead ECGs from nearly nineteen thousand patients, with diagnostic statements assigned by cardiologists. The high-resolution recordings, sampled at 500 Hz, were used throughout. Each record was resampled to a common 500 Hz rate where necessary and cropped or zero-padded to a fixed length of 5,000 samples, which corresponds to ten seconds of signal – long enough to contain several beats and therefore enough to judge rhythm.

The network is trained on two targets for each recording. The main objective is to obtain a single binary label, whereby the term AFIB is part of a diagnosis code, the recording is considered atrial-fibrillation-positive, and if it is not, then it is considered negative. The secondary target is a diagnostic superclass label, which is a kind of label derived from the five PTB-XL superclasses (NORM, MI, CD, HYP, and STTC) from the mapping of each recording’s SCP diagnostic statements to the superclass taxonomy. The superclass target is not mutually exclusive, but rather is considered to be multi-label because a single recording can include more than one diagnostic statement. This is why records with no valid superclass are excluded (Section 4.1): they don’t have a clearly defined superclass, and therefore cannot be inserted into the auxiliary task or imbalance-handling buckets, which rely on the superclass.

Splitting followed the dataset’s recommended stratified folds, with the ten predefined folds used as eight for training, the ninth for validation, and the tenth held out for testing. Keeping to this protocol matters: it respects the patient groupings, so that recordings from one patient do not leak across the train, validation, and test sets, and it keeps the results comparable to work that uses the same standard splits.

To test whether the detector generalises beyond the population and recording protocol it was trained on, a second, fully external dataset was used for evaluation only: the Chapman–Shaoxing twelve-lead ECG database [25]. This dataset was never seen during training or model selection, and it differs from PTB-XL in population and acquisition setup, making it a stricter test of generalisation than the held-out PTB-XL fold alone. The binary AF label for Chapman–Shaoxing was derived from the SNOMED-CT rhythm annotations supplied with the database: a recording was marked atrial-fibrillation-positive if the atrial-fibrillation code (SNOMED-CT 164889003) appeared among its diagnostic codes, and negative otherwise. Atrial flutter and all other rhythms were treated as negative, mirroring the AFIB-only definition used for PTB-XL. The external recordings were put through the same preprocessing pipeline described in Section 3.2.2 so that the comparison between the internal and external sets is fair.

3.2.2 Preprocessing and Augmentation

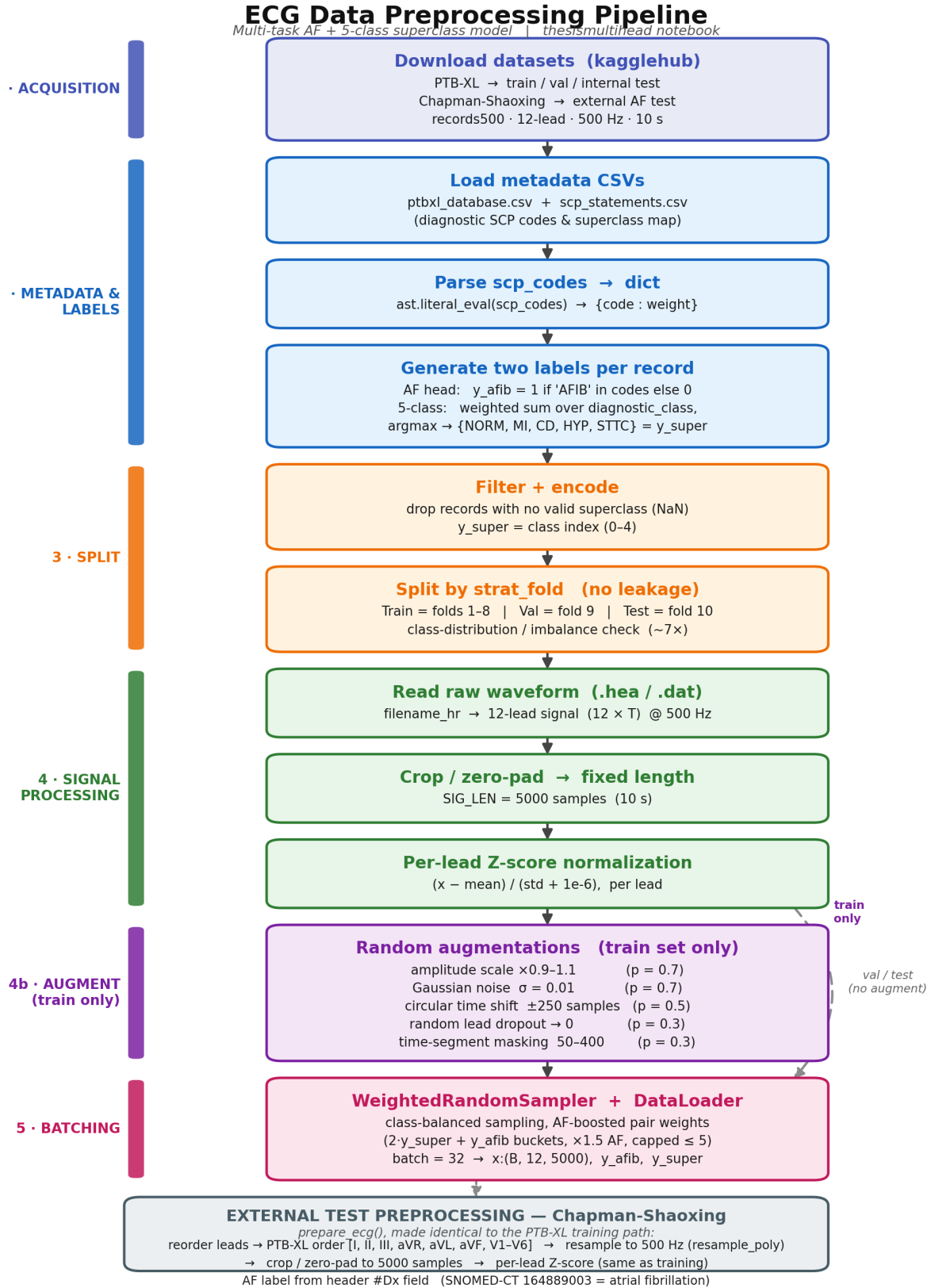


Figure 3.2: Preprocessing and augmentation pipeline.

Every recording passed through an identical preprocessing pipeline before reaching the model, and the same pipeline was applied to both the internal PTB-XL data and the external Chapman–Shaoxing data. Signals were taken at a 500 Hz sampling rate and fixed to a length of 5,000 samples (ten seconds). A high-pass filter with a cutoff of 0.5 Hz was applied to remove baseline wander and low-frequency drift, a step applied uniformly to both datasets. Each recording was then normalised lead by lead using a z-score, so that each channel had roughly zero mean and unit variance. Doing this per lead rather than globally keeps the relative information within a lead intact while removing the wide baseline differences between leads and between patients.

Because the training set is finite and the positive class is rare, the training signals were augmented on the fly [20] to discourage the model from memorising specific traces. Five transformations were applied stochastically: a small random amplitude scaling, additive Gaussian noise, a circular time shift, occasional dropping of an entire lead to zero, and the masking of a short random time window. Each was applied only some of the time, so the network sees a slightly different version of a recording on each pass. Validation and test signals were left untouched, augmentation being strictly a training-time device.

3.2.3 Handling Class Imbalance

Class imbalance was the single most influential design concern. Atrial fibrillation accounts for under 7% of the recordings, so a model trained naively on the natural distribution can reach a flattering accuracy simply by leaning toward the negative class, while quietly failing on the very cases the detector exists to catch. The imbalance was therefore countered on two fronts at once: at the sampling level, by reshaping the composition of each training batch, and at the loss level, by penalising errors on the rare class more heavily.

The sampling side used a weighted random sampler. Rather than weighting on the binary AF label alone, recordings were grouped into ten buckets formed by crossing the five diagnostic superclasses with the binary AF label, and a base weight inversely proportional to each bucket’s frequency was assigned, so that rare combinations were drawn more often than common ones. These weights were then normalised to unit mean. Two adjustments tempered the scheme: a mild multiplicative boost of 1.5 was applied to the five AF-positive buckets to lift the minority class further, and every weight was capped at a maximum of 5.0 to stop the smallest buckets from dominating a batch outright. Sampling was performed with replacement over the resulting per-recording weights.

On the loss side, the binary cross-entropy objective was weighted in favour of the positive class, so that a missed AFIB case incurred a larger penalty than a false alarm. This complements the sampler: where the sampler controls how often the model sees the rare class, the weighted loss controls how much each such example counts once seen. Together they push the model toward the recall-oriented behaviour appropriate to a screening role.

The combined effect, examined in Chapter 4, is that the rare positive class is lifted to a substantial share of training signal rather than being drowned out, which is the precondition for the model to learn it at all.

3.2.4 Model Architecture

The network is a one-dimensional convolutional encoder followed by a bidirectional LSTM and a compact detection head, taking the twelve-lead signal as input and producing a single logit for the atrial-fibrillation decision. Table 3.1 summarises the layer configuration.

The encoder is a stack of convolutional blocks, each pairing a one-dimensional convolution with batch normalisation [32], a GELU activation [31], and dropout [37] of 0.15. The twelve input leads are first widened to 64 channels by a kernel-9, stride-2 convolution and refined by a kernel-7 block, after which a max-pooling layer halves the time dimension. The channel width is then raised to 128 through a strided kernel-7 block and a kernel-5 block, followed by a second max-pooling layer, and finally to 192 through a strided kernel-5 block and a kernel-3 block. The net effect is that each lead is distilled into a much shorter sequence of 192-dimensional feature vectors, with the strided convolutions and pooling progressively trading time resolution for representational depth.

This feature sequence is passed to a two-layer bidirectional LSTM with a hidden size of 128 per direction. Reading the sequence in both directions, it produces at each step a 256-dimensional feature reflecting both past and future context. The per-step outputs are averaged over time into a single 256-dimensional vector summarising the whole recording, and a compact head – a linear layer to 128 units, a GELU activation, dropout of 0.4, and a final linear layer – maps that vector to the single atrial-fibrillation logit.

3.3 Design Considerations

3.3.1 Public Health and Safety Considerations

A model of this kind is a decision-support aid, not a replacement for clinical judgement, and the design reflects that. The deliberate effort to detect the rare positive class is in part a safety consideration: in screening, a missed arrhythmia is far more costly than a false alarm, which is why both the sampling and the loss were tilted toward the minority class even at the expense of some additional false positives. The same priority is visible in how the operating threshold was chosen – on the validation set rather than fixed at 0.5 – so that the recall-oriented behaviour appropriate to screening was set deliberately rather than left to chance. The explanation component, described below, exists for the same reason: a clinician can only sensibly act on a prediction whose basis they can inspect. No claim is made that the system is fit for unsupervised clinical deployment; it is intended to be used under expert oversight.

Table 3.1: Layer configuration of the CNN–BiLSTM detector. Each ConvBlock is a 1-D convolution followed by batch normalisation, GELU, and dropout (0.15). Kernel (k), stride (s), and padding (p) are given per convolution.

Stage	Layer	Output channels
Input	12-lead ECG (12×5000)	12
Encoder	ConvBlock ($k=9, s=2, p=4$)	64
	ConvBlock ($k=7, s=1, p=3$)	64
	MaxPool ($/2$)	64
	ConvBlock ($k=7, s=2, p=3$)	128
	ConvBlock ($k=5, s=1, p=2$)	128
	MaxPool ($/2$)	128
	ConvBlock ($k=5, s=2, p=2$)	192
	ConvBlock ($k=3, s=1, p=1$)	192
Recurrent	BiLSTM (2 layers, hidden 128/direction)	256
Pooling	Mean over time	256
Head	Linear $\rightarrow$ GELU $\rightarrow$ Dropout (0.4)	128
	Linear	1

3.3.2 Cultural, Societal, and Environmental Considerations

Because the model is trained on a single dataset gathered under one recording protocol, its behaviour reflects the population and equipment behind that data. This is a fairness consideration as much as a technical one, since performance need not carry over to patient groups or devices unlike those in the training data. The thesis takes this concern seriously rather than implying broad generalisation, and it was precisely this worry that motivated evaluating the trained detector on a second, independent dataset – the Chapman–Shaoxing database – drawn from a different population and acquisition setup, so that the question of transfer is examined empirically rather than assumed. On the environmental side, the network was kept deliberately compact, and the explanation step reuses a frozen pre-trained model through prompting rather than any further training, both of which hold down the computational cost of building and running the system.

3.4 Simulation and/or Experimental Methods

3.4.1 Training Procedure

Training was carried out in PyTorch [29]. The detector was optimised to minimise a binary cross-entropy loss on the atrial-fibrillation label, weighted in favour of the positive class as described in Section 3.2.3, with the weighted sampler supplying rebalanced batches. Optimisation used AdamW [28] with a learning rate of 2×10^{-3} and weight decay of 1×10^{-4} ,

on mini-batches of 32 recordings. The learning rate was reduced automatically whenever validation performance plateaued, gradient norms were clipped [38] to keep updates stable, and mixed-precision arithmetic [33] was used on the GPU to save memory and time. Training ran for up to 35 epochs with early stopping, halting once validation performance had not improved for 8 consecutive epochs, and the best checkpoint – selected on validation performance rather than training loss – was retained for all subsequent evaluation. Selecting on validation in this way guards against the model continuing to fit the training data after its generalisation had stopped improving. A fixed random seed was used to aid reproducibility. The detailed epoch-by-epoch behaviour, and the point at which the best checkpoint was taken, are presented in Chapter 4.

Once trained, the model was frozen and evaluated in two settings. The internal test used the held-out PTB-XL fold; the external test used the Chapman–Shaoxing dataset, with no parameter updates of any kind, so that it functions as a genuine test of generalisation rather than a second training stage. The decision threshold was not left at the default of 0.5 but chosen on the validation set by maximising Youden’s J statistic [27], which balances sensitivity against specificity at a single operating point. This validation-derived threshold was then applied unchanged to both test sets; for the external set, results at the default threshold of 0.5 are also reported, so that the model’s behaviour can be read both at its tuned operating point and at a neutral one.

3.4.2 Explainability and Explanation Generation

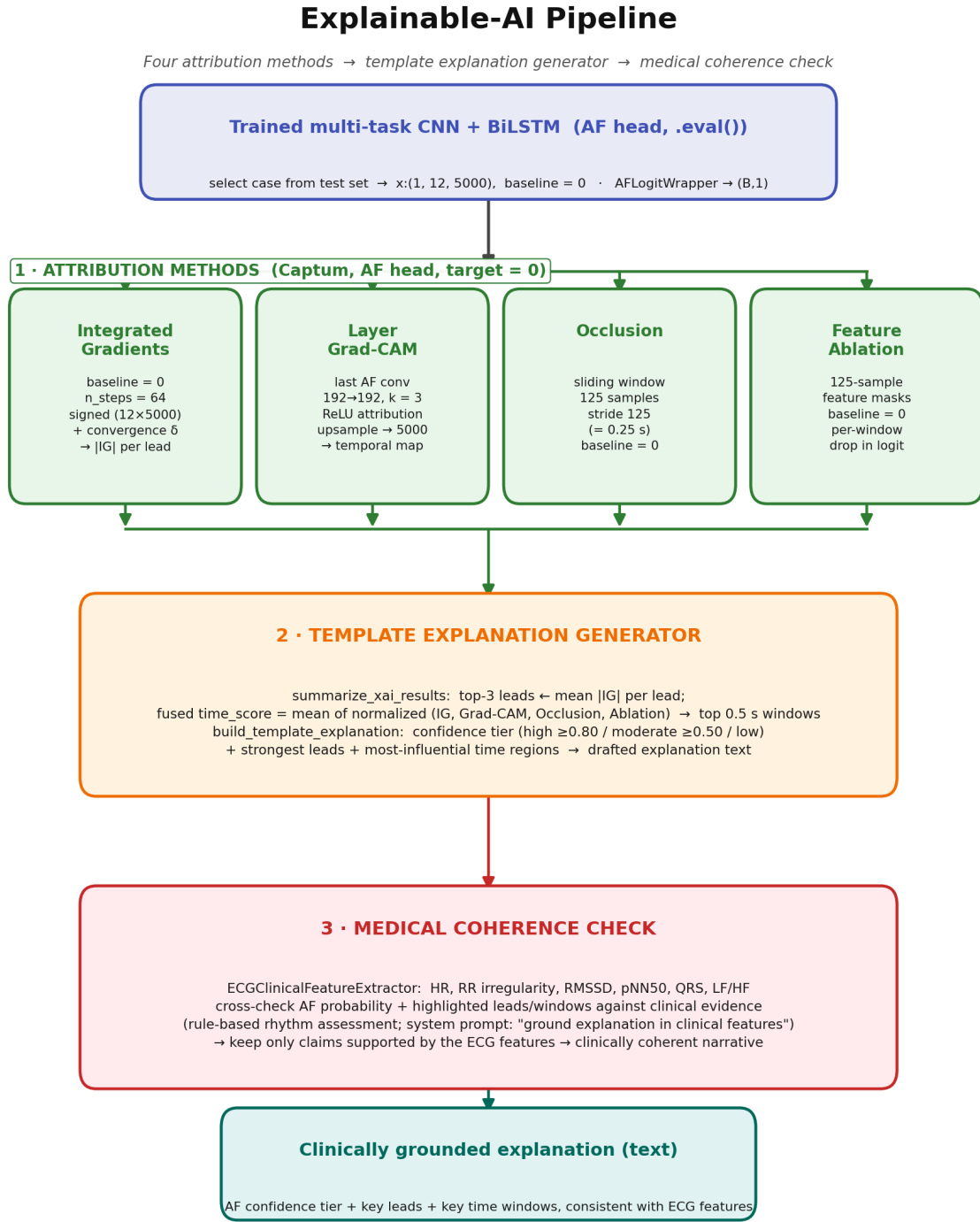


Figure 3.3: Overview of the explainability pipeline

Interpretability was treated as part of the method, not an afterthought. Four attribution techniques from the Captum library [26] were applied to the trained model. Integrated Gradients traces a prediction back to input features by accumulating gradients along a path from a neutral baseline. Grad-CAM uses the gradients within a convolutional layer to localise

the regions that drove the decision. Occlusion and feature ablation, two perturbation-based methods, measure how the prediction changes as parts of the signal are masked out or removed. Run together on the same recording, the four offer cross-checking views rather than a single perspective, and their agreement – or disagreement – becomes a way of judging how much weight any one map deserves. The raw attributions were then condensed into something a reader can use: the leads carrying the most attribution and the half-second time windows where attribution concentrated, turning a dense heatmap into a short list of where the model was looking.

These condensed findings were then rendered into a brief natural-language explanation. Rather than train a dedicated text model, the system reuses a frozen pre-trained language model and supplies it with a structured prompt assembled from three sources: the model’s own predictions (the AFIB probability and the five-class superclass probabilities), a set of clinically interpretable signal features computed from the raw recording, and the attribution summary (the leads carrying the most attribution and the most influential time windows). The signal features are heart-rate and heart-rate-variability measures derived from Lead II R-peak detection— mean and standard deviation of the RR intervals, RMSSD, pNN50, and an RR-irregularity score defined as the coefficient of variation of the RR intervals, the last being a direct marker of the irregular ventricular response characteristic of atrial fibrillation — together with the LF/HF spectral-power ratio and per-lead amplitude ranges. A rule-based rhythm interpretation derived from the irregularity score and heart rate is also included. The language model is then instructed to phrase these findings as a short, readable explanation. The template fixes the factual content, drawn directly from the attribution analysis, while the language model supplies fluent wording, so that the explanation reflects the model’s evidence rather than free-form generation. Figure 3.4 shows a representative output for a correctly detected case, in which the named leads and time windows match those surfaced by the attribution methods.

Figure 3.4: Example natural-language explanation for a correctly detected AFIB case, generated from the predicted probability and the leads and time windows identified by the attribution methods.

3.4.3 Limits of the Methods

As required of any data-driven study, the boundaries of these tools should be stated plainly. Attribution methods indicate where a model attends, not whether its reasoning is clinically correct, and the methods used here can disagree at the margins. The generated explanations inherit this limitation: they faithfully report what the attribution analysis found, but that is a statement about the model’s behaviour, not an independent clinical judgement, and they have not been validated against expert annotation. The fixed ten-second window discards anything outside it. External validation on the Chapman–Shaoxing dataset tempers, but does

not remove, the concern that findings rest on a narrow recording protocol: two datasets are a stronger basis than one, yet still fall short of the range of devices and populations a deployed system would meet. These limits inform the results in Chapter 4 and the directions raised in Chapter 5.

CHAPTER 4:

RESULTS AND DISCUSSION

This chapter reports what the detector achieved and tries to make sense of it. It opens with the data the experiments were built on, since the shape of that data explains a good deal of what follows. Then turns to how training progressed, how the model performed on the held-out test set. And finally what the explainability analysis revealed about the features the network relied on. Where a result is encouraging it is stated plainly; where it is mixed, that is reported just as plainly. Because a result section that hides its weak spots is of little use to anyone trying to build on it.

4.1 Dataset Composition

The experiments draw on two datasets: PTB-XL [6] for training, validation, and internal testing, and the Chapman–Shaoxing database [25] for external testing only. Table 4.1 summarises both.

PTB-XL contains 21,837 twelve-lead recordings in total. Records lacking a clear diagnostic superclass label were excluded, leaving a working set of 21,417. This step is not strictly required for the binary AFIB task, but it was retained so that every recording carries a well-defined superclass -which the imbalance-handling scheme depends on; since the weighted sampler buckets recordings by crossing the superclass label with the AFIB label (Section 3.2.3). A record with no superclass could not be placed in that scheme, so removing the 420 such records keeps the labelled subset consistent and well-characterised.

Following the dataset’s predefined folds, the working set was divided into 17,100 recordings for training, 2,155 for validation, and 2,162 held out for testing. With the proportion of atrial fibrillation reproduced closely across all three splits - which is precisely the point of using the stratified folds. The imbalance that shaped so much of the methodology is plain in the numbers. Atrial fibrillation is present in only 1,157 of the 17,100 training recordings, or 6.77%, with the validation and test folds carrying a near-identical 6.82% (147 of 2,155) and 6.66% (144 of 2,162). In other words, roughly fourteen of every fifteen recordings are negative.

It is worth pausing on what this means in practice. A naive classifier could reach about 93% accuracy by always predicting “not AFIB” and would still catch none of the arrhythmia cases- exactly the cases the detector exists to flag. This is why the discussion that follows leans on recall, precision, F1, and ROC-AUC for the positive class rather than overall accuracy, which a model can inflate simply by learning the base rate.

The external Chapman-Shaoxing set differs from PTB-XL in a way that matters for interpreting the results later. It contributes 10,247 recordings for evaluation, of which 1,780 are AFIB-positive - a prevalence of 17.4%, more than double PTB-XL’s rate. Because precision depends on how common the positive class is, this higher base rate must be kept in mind when comparing the internal and external numbers; a model can achieve higher precision on Chapman partly because positives are simply less rare there.

Table 4.1: Composition of the datasets used for training and evaluation. PTB-XL follows the predefined stratified folds; Chapman–Shaoxing is used for external testing only.

Dataset	Split	Records	AFIB	AFIB %
PTB-XL	Train	17,100	1,157	6.77
	Validation	2,155	147	6.82
	Test	2,162	144	6.66
Chapman–Shaoxing	External	10,247	1,780	17.4

4.2 Effect of the Balanced Sampler

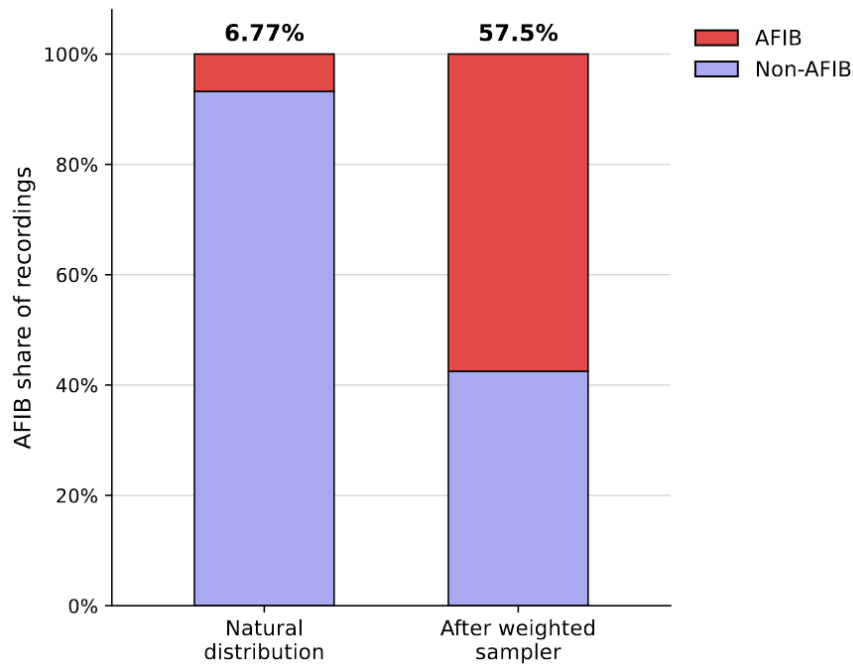


Figure 4.1: Effect of the weighted random sampler on class balance. In the natural PTB-XL training distribution, atrial fibrillation accounts for only 6.77% of recordings (left); after weighted sampling, AFIB-positive examples make up roughly 57–58% of each training batch (right), close to a balanced batch.

Before training, it was worth checking that the weighted sampler actually did what it was meant to. Drawn naively, a typical batch would hold only one or two atrial fibrillation examples, leaving the model almost nothing positive to learn from on most steps. The weighted sampler changed this substantially. Its effect is visible in the assigned weights: the scarcest groups of positive recordings were lifted to the imposed ceiling of 5.0. While the abundant negative recordings were pushed down to as little as 0.024. This spread is the combined product of the inverse-frequency base weighting, the mild $1.5\times$ boost applied to the AFIB-positive buckets, and the cap that prevents the smallest buckets from dominating outright, all described in Section 3.2.3.

When batches were then sampled and inspected, the proportion of AFIB-positive examples rose to roughly 0.57–0.58 on average across repeated checks; close to a balanced batch, against a natural rate below 7%. That the figure lands near one-half is by design: the boost and cap were chosen so that positives are well represented without the sampler tipping into the opposite failure of flooding every batch with the same handful of recordings. Because sampling is done with replacement over 1,157 distinct positive recordings, this rate still draws on the great majority of them across an epoch rather than recycling a narrow few. So the gain in exposure does not come at the cost of overfitting to a small positive subset. In short, the rare positive class was no longer drowned out at the batch level, which is the precondition for the model to learn it at all. This is a means-to-an-end result rather than a headline one. But it confirms the rebalancing behaved as intended.

4.3 Training Behaviour

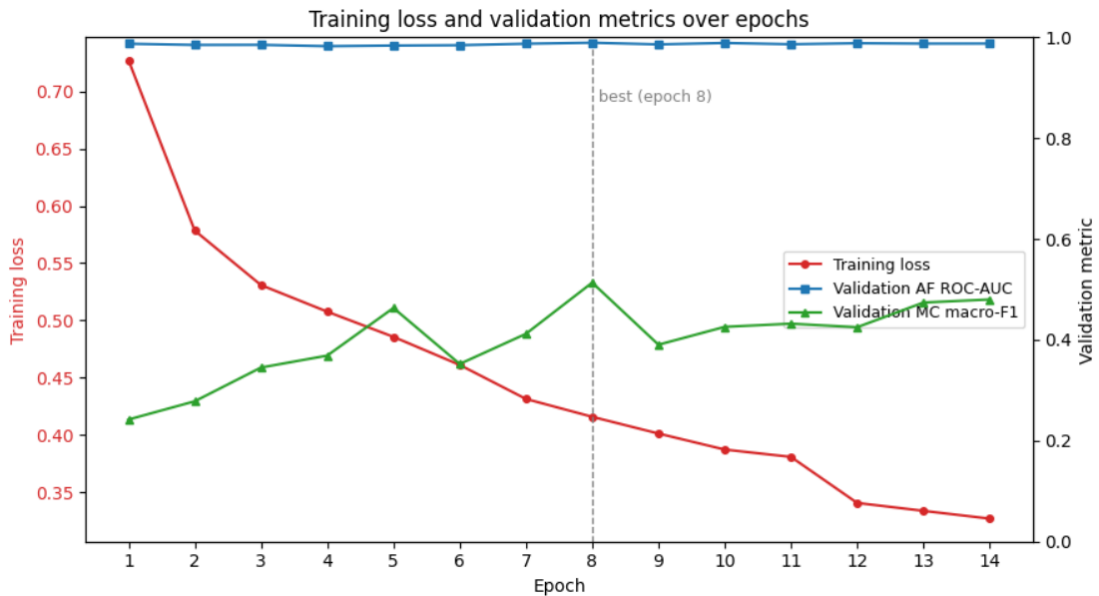


Figure 4.2: Training loss together with the validation atrial fibrillation ROC-AUC and F1 across the fourteen training epochs.

The detector learned its task quickly and then refined it. The validation ROC-AUC for atrial fibrillation sat near 0.98 from the very first epoch and never left that band, peaking around 0.989. This is less surprising than it looks. Atrial fibrillation carries a distinctive signature of irregular R–R intervals and absent P-waves. Which the network can latch onto quickly even from relatively few positive examples, helped along by the oversampling. The validation F1 for the positive class climbed more gradually as the decision boundary sharpened, rising from about 0.71 in the first epoch into the mid-0.80s, while the training loss fell steadily throughout, from about 0.73 down to 0.33.

Model selection was tied to validation performance rather than training loss. So that the run would not simply keep fitting the training data after its validation behaviour had plateaued. The best validation checkpoint was reached at epoch 8, where the AFIB ROC-AUC was about 0.989 and the positive-class F1 was among the strongest of the run. Training was allowed to continue but never beat that checkpoint, and early stopping ended the run at epoch 14. All test results reported below come from the epoch-8 model.

4.4 Test-Set Performance

The decisive numbers are those on the held-out test fold, which the epoch-8 model never saw during training or selection. Table 4.2 collects the metrics for the positive (AFIB) class alongside the overall figures.

Table 4.2: Atrial fibrillation detection performance on the held-out PTB-XL test set (fold 10), epoch-8 model, at the validation-derived operating threshold.

Metric	Value
ROC-AUC	0.9791
F1-score (AFIB class)	0.8272
Precision (AFIB class)	0.7444
Recall (AFIB class)	0.9306
Precision (non-AFIB class)	0.9950
Recall (non-AFIB class)	0.9772
F1-score (non-AFIB class)	0.9860
Overall accuracy	0.9741
Macro-average F1	0.9066

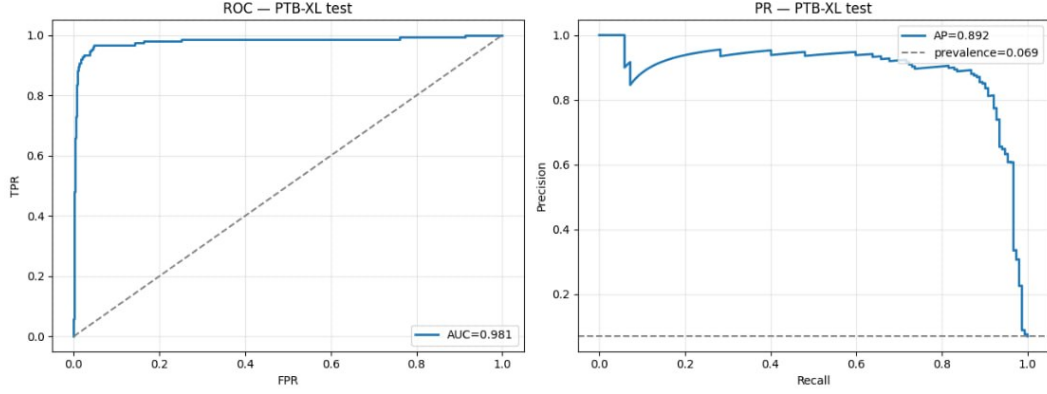


Figure 4.3: ROC and precision–recall curves for the held-out PTB-XL test fold.

The detector performed well. It reached a test ROC-AUC of 0.979 and, at the operating threshold selected on the validation set by maximising Youden’s J , recalled 93.1% of the true AFIB cases. The precision on the AFIB class was lower at 0.744, meaning roughly a quarter of its positive calls were false alarms. For a screening setting this is a defensible trade. Catching better than nine in ten arrhythmia cases at the cost of some over-flagging is usually preferable to the reverse, since a missed arrhythmia is the more dangerous error. The negative class, predictably, was classified almost perfectly given its abundance, with an F1 of 0.986, and the two together give a macro-averaged F1 of 0.907- a more honest summary than the headline accuracy of 0.974, which the class balance alone would push upward.

It is worth dwelling briefly on the precision–recall balance, since it is the main lever a deployer would adjust. The model was tuned to favour recall, on the view that for a screening aid a false negative- a missed arrhythmia- is worse than a false positive. The 0.744 precision reflects that deliberate choice rather than a failure of the model. Moving the decision threshold upward would trade some recall for higher precision, and the right operating point depends on how the tool is used. Reported at this validation-derived operating threshold, the balance struck here is appropriate for a first-pass screen that flags candidates for expert review.

4.5 Diagnostic Superclass Classification

The auxiliary diagnostic head was evaluated on the same held-out PTB-XL test fold as the binary detector, using the frozen epoch-8 model. Because a recording may carry several diagnostic statements, the task is multi-label. Each of the five superclasses is predicted independently with a sigmoid output, and point metrics are reported at a decision threshold of 0.5.

```

MC superclass – MACRO-AUC = 0.9083
NORM  AUC = 0.9403
MI     AUC = 0.9194
STTC   AUC = 0.9279
CD     AUC = 0.9265
HYP    AUC = 0.8275

```

Figure 4.4: Dignostic superclass classification

4.6 External Validation on Chapman–Shaoxing

A model that performs well on a held-out fold of the data it was trained on has shown only that it generalises within one dataset and one recording protocol. The stricter question-whether it transfers to recordings from a different population and acquisition setup - can only be answered on data the model has never encountered. To test this, the epoch-8 model was applied, frozen and unchanged, to the Chapman–Shaoxing database. No retraining, fine-tuning, or threshold refitting on Chapman took place; the preprocessing pipeline was identical to that used for PTB-XL, and the operating threshold carried over directly from the PTB-XL validation set. Table 4.3 reports the results at two thresholds: the validation-derived Youden’s J threshold used internally, and the neutral default of 0.5.

Table 4.3: External validation on the Chapman–Shaoxing dataset (10,247 recordings, frozen epoch-8 model). Results are shown at the PTB-XL validation-derived threshold and at the default threshold of 0.5.

Metric	Val-threshold	Threshold 0.5
ROC-AUC	0.9610	0.9610
Precision (AFIB class)	0.552	0.602
Recall (AFIB class)	0.973	0.942
F1-score (AFIB class)	0.704	0.734
Precision (non-AFIB class)	0.993	0.986
Recall (non-AFIB class)	0.834	0.869
F1-score (non-AFIB class)	0.907	0.924
Overall accuracy	0.858	0.882
Macro-average F1	0.805	0.829

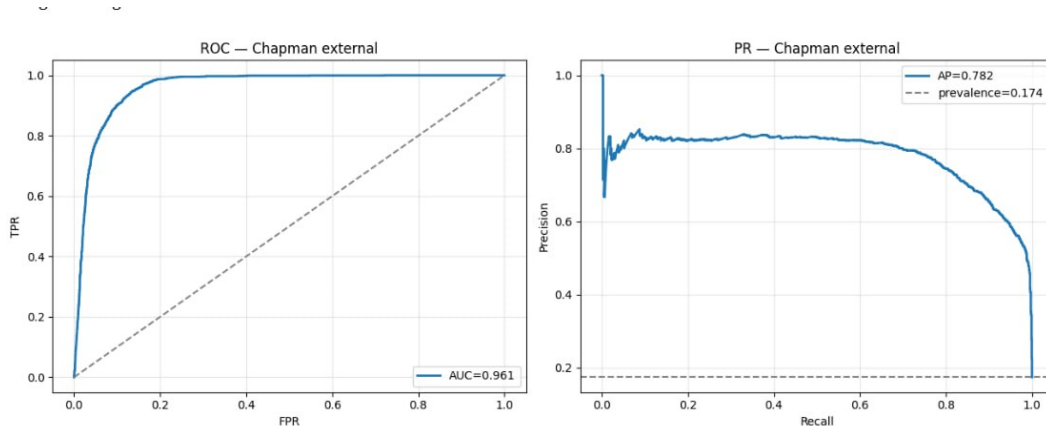


Figure 4.5: ROC and precision–recall curves for the external Chapman–Shaoxing evaluation.

The headline result is that the detector held up well on a genuinely unseen dataset. The ROC-AUC of 0.961 is only modestly below the internal 0.979, which is reassuring: the model’s ability to rank AFIB above non-AFIB recordings carried over to a new population with little loss. Recall on the positive class actually rose, to 0.973 at the validation threshold, so the model continued to catch the great majority of true arrhythmia cases- exactly the behaviour a screening tool needs to preserve when moved to new data.

The cost of that transfer shows up in precision, which fell to 0.552 at the validation threshold and 0.602 at the default. In other words, the model over-flagged more on Chapman than on PTB-XL. Part of this is the expected behaviour of any fixed model on out-of-distribution data, but part is explained by the difference in prevalence noted in Section 4.1. Although Chapman’s AFIB rate of 17.4% is higher than PTB-XL’s, the model’s threshold was tuned on PTB-XL’s distribution, and a threshold carried unchanged to a different population will rarely sit at the ideal operating point. This is visible in how the two thresholds trade off: raising the threshold from the validation value toward 0.5 recovered some precision (0.552 to 0.602) at a small cost in recall (0.973 to 0.942). which is precisely the lever discussed for the internal results and confirms that the operating point, not the model’s discriminative power, drives the precision figure.

Read together, the internal and external results tell a consistent story. The model’s ranking ability- captured by ROC-AUC - is strong and stable across both datasets, and its recall remains high throughout, so it rarely misses a true arrhythmia. The precision it achieves depends heavily on the operating threshold and the prevalence of the population it is applied to, and a threshold tuned on one dataset does not transfer perfectly to another. For a screening aid this is an acceptable and well-understood trade, but it also points clearly to a practical lesson: a deployment on a new population would benefit from recalibrating the threshold on a small sample of that population’s data, rather than carrying the original threshold across unchanged.

It is worth placing this result against the published literature, though such comparisons have to be read with care. Studies differ in how they frame the task, how they split the data, and whether they report a single binary score or a macro-average over many labels. Table 4.4 sets the present result beside the most authoritative PTB-XL benchmark, and the qualifications matter as much as the numbers.

Table 4.4: Reported PTB-XL rhythm-classification performance from the standard benchmark, alongside the present work. The tasks differ in framing and evaluation, so the figures are indicative rather than strictly comparable; see the discussion below.

Study / model	Task	AUC
Strodthoff et al. [7] (xres-net1d101)	Rhythm category, multi-label macro-AUC	0.957
Strodthoff et al. [7] (bidirectional LSTM)	Rhythm category, multi-label macro-AUC	0.953
This work	Binary AFIB detection (raw signal)	0.979

The Strodthoff benchmark figures are macro-averaged AUC across the whole *rhythm* category, a multi-label problem spanning several rhythm types, rather than a single binary atrial-fibrillation score. So they measure a broader task than the one here. A dedicated binary detector can be expected to score higher on its single target than a model splitting its capacity across many labels. so the gap is not by itself evidence of a better method. It is reassuring that the present figure lands close to and slightly above the strongest single-model rhythm benchmark on the same dataset and the same fold-based protocol. Which at least places the detector in credible company. Other work has reported binary atrial-fibrillation AUCs on PTB-XL in roughly the low-0.93 range [35] for raw-signal models and near-perfect values for image-based approaches [35] under different train–test splits; the latter are not comparable to the raw-signal, fold-based setup used here. The fair conclusion is that this result sits well within the range of credible PTB-XL performance for atrial-fibrillation detection, without claiming a new state of the art, a claim that would require a head-to-head comparison on an identical protocol to support.

4.7 Explainability Results

A correct prediction is more trustworthy when its basis can be inspected, and this is what the attribution analysis was for. Applying the attribution methods to test recordings produced, for each case, a map over the twelve leads and the ten-second window showing which parts of the signal most influenced the decision.

4.7.1 Case Studies

To keep the analysis concrete, two representative recordings were carried through the pipeline: an atrial-fibrillation-positive case (record 16487) and an atrial-fibrillation-negative case (record 1263). Figures 4.6 and 4.7 show the full twelve-lead traces for each, and Figure 4.8 places their Lead II signals side by side, Lead II being the conventional choice for inspecting rhythm.

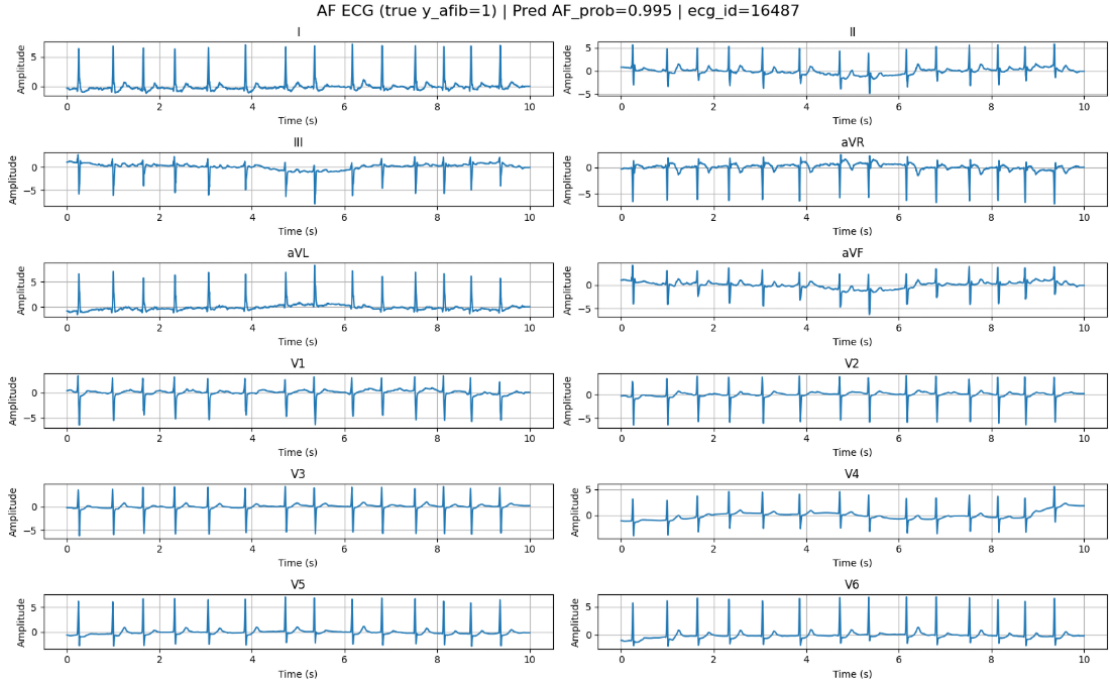


Figure 4.6: Twelve-lead trace of the atrial-fibrillation-positive case (record 16487), correctly detected with an AFIB probability of 0.995

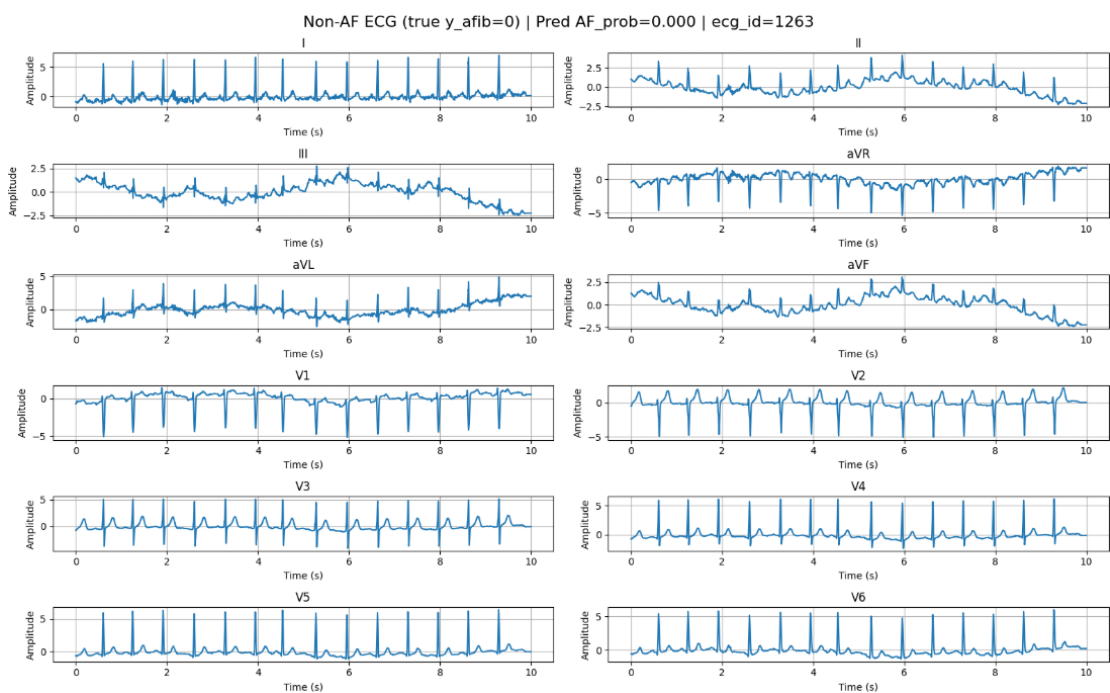


Figure 4.7: Twelve-lead trace of the atrial-fibrillation-negative case (record 1263), correctly rejected with an AFIB probability of 0.000.

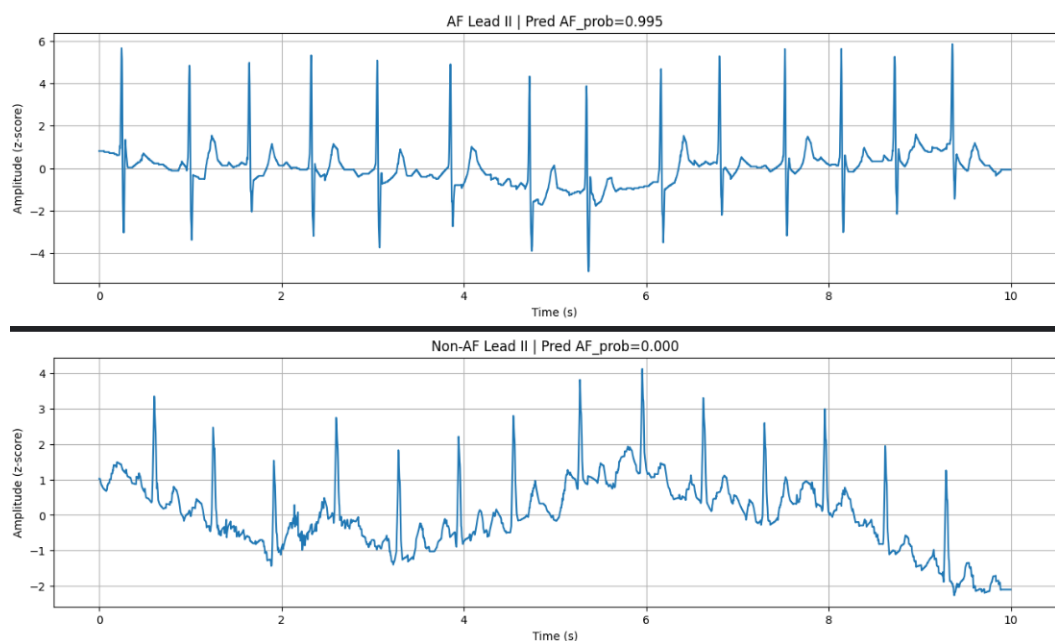


Figure 4.8: Lead II comparison of the two case-study recordings: the AFIB-positive record 16487 (top) and the AFIB-negative record 1263 (bottom).

It is worth being candid about what these raw traces do and do not show, since a reader will look at them directly. The two recordings differ visibly in baseline stability – the AFIB-negative trace carries noticeably more low-frequency baseline wander – but the rhythm irregularity that defines atrial fibrillation is not dramatically obvious to the eye in this particular

pair. This is part of why an automated detector is useful. The model’s decision rests on subtler beat-to-beat timing and morphology cues than a quick glance at a single lead reveals, which is exactly the kind of evidence the attribution analysis is meant to surface.

On these two cases the detector was decisive and correct. It assigned the AFIB-positive record an AFIB probability of 0.995 and the AFIB-negative record a probability of essentially zero, matching their true labels in both cases. These are the kind of confident, correct calls the test-set numbers describe in aggregate, and they make a clean pair of examples to carry into the attribution analysis that follows.

4.7.2 Attribution Maps

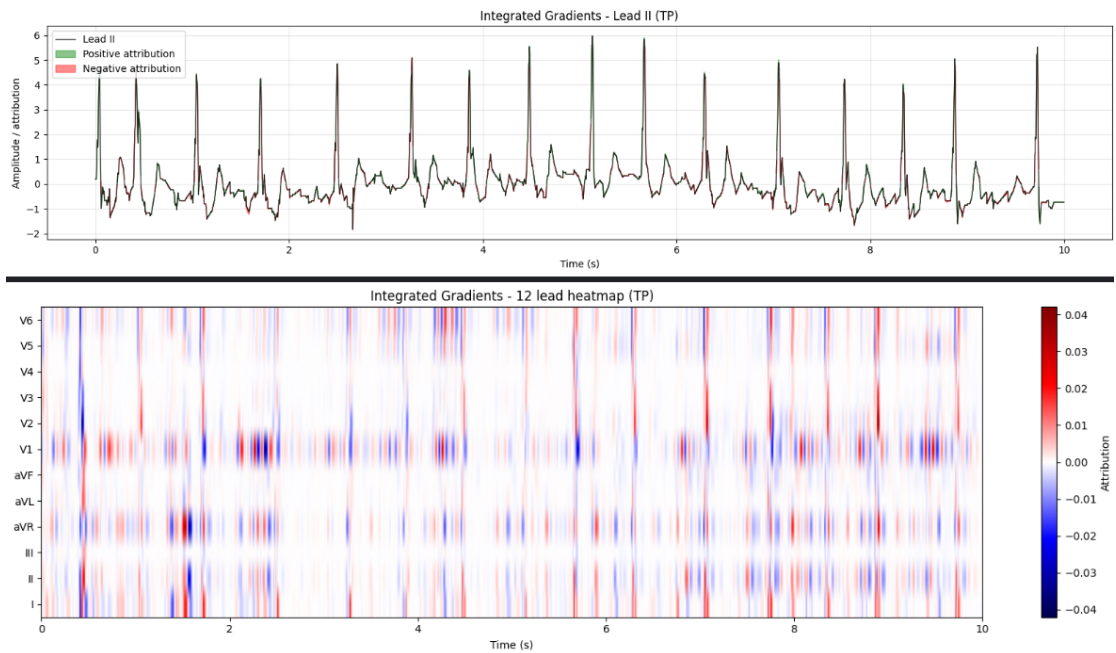


Figure 4.9: Integrated Gradients attribution for a correctly detected atrial fibrillation case (test sample 42, predicted AFIB probability 0.989). Top: Lead II with positive and negative attribution overlaid on the trace. Bottom: attribution across all twelve leads, where red and blue indicate the sign of the contribution.

Figure 4.9 shows the Integrated Gradients result for a true-positive AFIB case – test sample 42, which the model flagged with an AFIB probability of 0.989. Two things are worth reading from it. The first is which leads mattered most. Ranking the leads by their mean absolute attribution put V1 at the top (0.0038), followed by aVR (0.0028), Lead II (0.0022), Lead I (0.0019), and V6 (0.0018). This ordering is encouraging from a clinical standpoint: V1 is the lead conventionally used to assess atrial activity and is where fibrillatory waves tend to be most visible, while Lead II is the standard rhythm-monitoring lead. That the model’s attribution falls most heavily on exactly the leads a clinician would inspect for an

atrial rhythm question suggests it is drawing on physiologically relevant regions rather than incidental ones.

The second is where in time the attribution concentrates. Both the Lead II overlay and the twelve-lead heatmap show attribution clustering around the QRS complexes - the sharp beat-to-beat deflections- rather than spreading evenly across the trace. Scoring the signal in half-second windows identified the most influential intervals as roughly 4.00–4.50 s, 8.50–9.00 s, and 7.50–8.00 s, each coinciding with beat activity rather than quiescent stretches. For a rhythm-based decision this is the sensible place for the model to look, since the timing and morphology of successive beats is precisely what distinguishes a fibrillating rhythm. It should be said plainly, though, that an attribution map establishes where the model attended, not that its internal reasoning is clinically correct; the alignment with rhythm-relevant leads is supporting evidence, not proof. The Integrated Gradients convergence delta for this case was moderate (about -0.23), which indicates the attribution is reasonable but not perfectly converged, and is worth reporting honestly rather than omitting.

Because the analysis also applied Grad-CAM and two perturbation-based methods - occlusion and feature ablation- alongside Integrated Gradients. The degree to which they agree is itself a test of how much the explanation can be trusted. They agree closely. The perturbation methods, which work by masking or removing parts of the signal and measuring the effect on the prediction, both ranked the leads V1 (0.093), aVR (0.066), V2 (0.059), Lead I (0.058), and V6 (0.057) – placing V1 first and aVR second, exactly as Integrated Gradients did, and sharing four of its top five leads (V1, aVR, I, and V6), with the only swap being V2 in place of Lead II at third. Grad-CAM, applied to the final convolutional layer and upsampled back to the full signal length, independently produced a temporal importance profile whose peaks line up with the individual beats, echoing the QRS-aligned concentration seen in the other maps.

Figures 4.10 and 4.11 show the Grad-CAM temporal profile and a perturbation-based attribution respectively. The feature-ablation map (Figure 4.12) tells the same story through a third mechanism: removing the segments around the QRS complexes produces the largest swing in the prediction, and the per-lead panel again concentrates on V1 and aVR, in agreement with the integrated gradients and occlusion results. The significance of this agreement is that the methods rest on entirely different principles- Integrated Gradients on accumulated gradients, the occlusion and ablation methods on input perturbation, and Grad-CAM on convolutional-layer activations. Yet they converge on the same small set of leads and the same beat-aligned timing. When independent techniques point to the same evidence, it is far less likely that the highlighted regions are an artefact of any one method, and far more likely that they reflect something the model genuinely relies on. For an atrial fibrillation decision resting most heavily on V1 and aVR, that is a reassuring outcome.

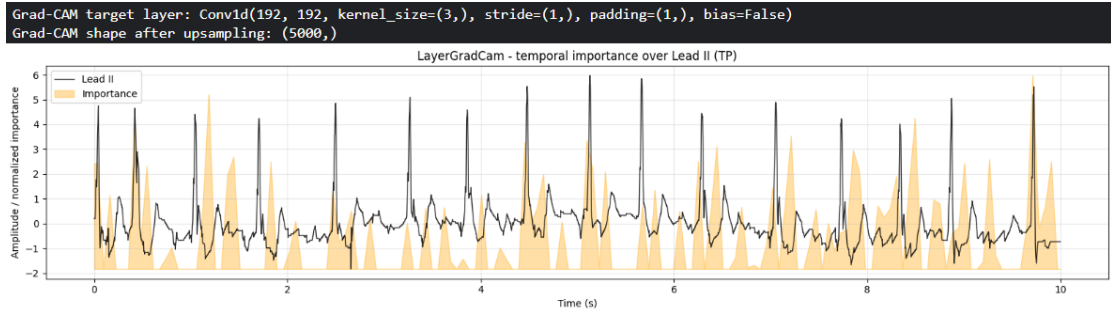


Figure 4.10: Grad-CAM temporal importance over Lead II for the same true-positive case, computed at the final convolutional layer and upsampled to the full signal length. Shaded importance peaks align with the individual beats.

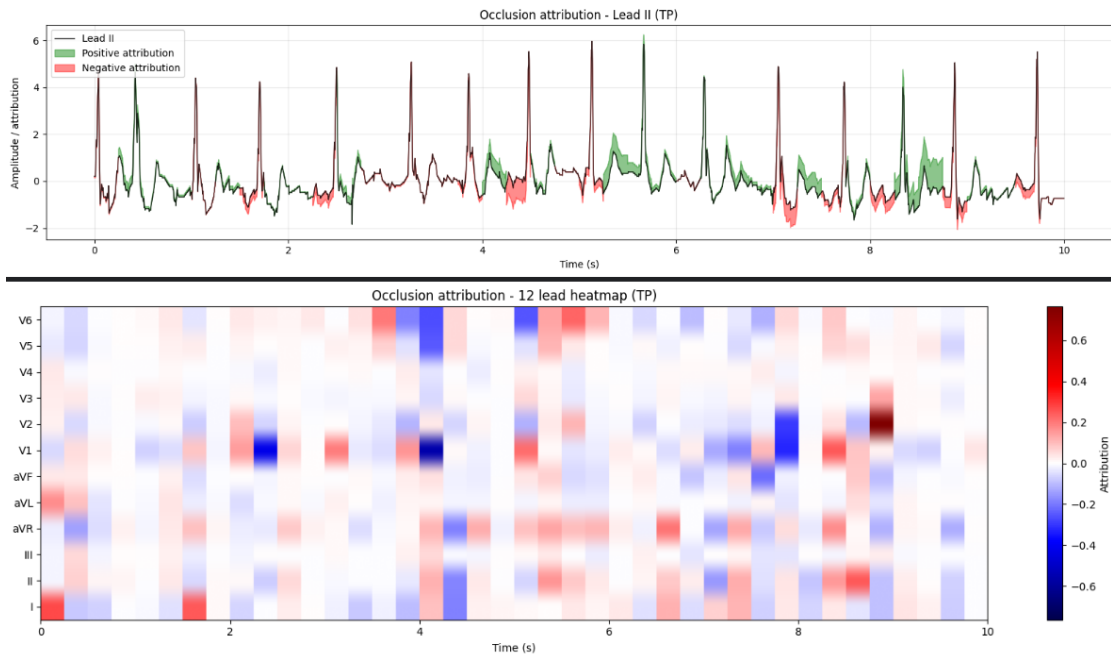


Figure 4.11: Occlusion attribution for the same true-positive case. Top: Lead II with positive and negative attribution overlaid. Bottom: attribution across all twelve leads, again concentrated on V1 and aVR.

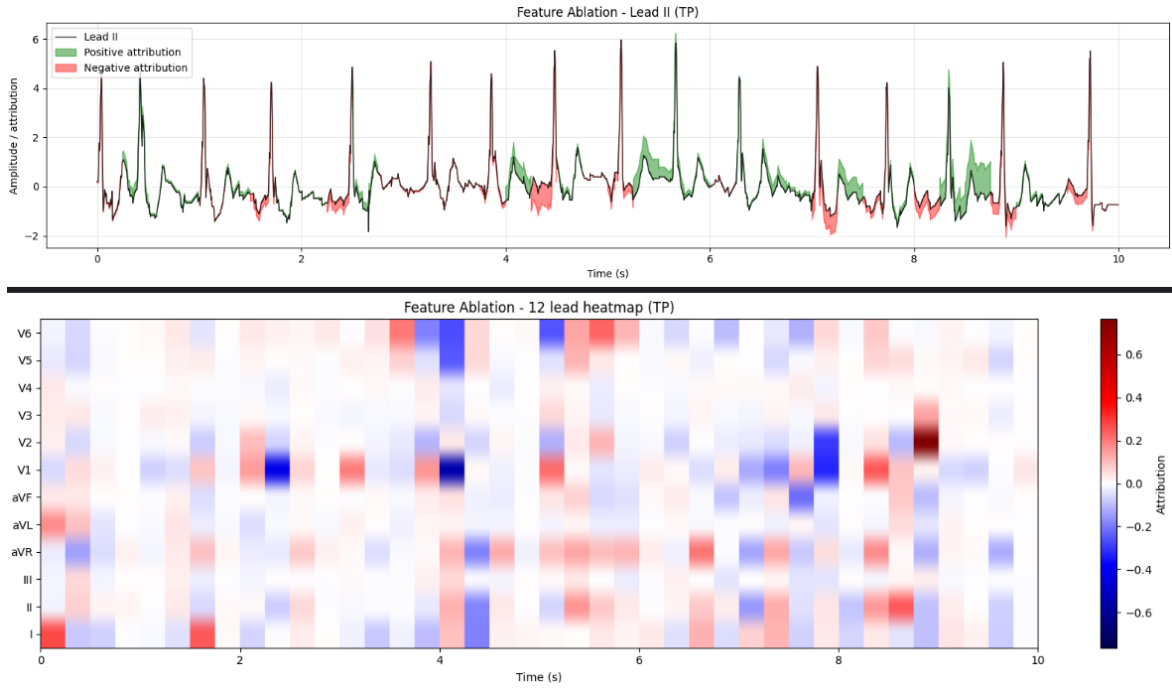


Figure 4.12: Feature ablation attribution for the same true-positive case. Top: Lead II with positive (green) and negative (red) attribution overlaid on the trace. Bottom: attribution across all twelve leads, with the colour scale indicating the sign and magnitude of each lead’s contribution.

4.7.3 Global Lead Importance

A single well-behaved example is suggestive but not conclusive. It could always be the one case that happens to look tidy. To test whether the pattern held more broadly, Integrated Gradients was averaged over twenty atrial-fibrillation-positive recordings to give a population-level view of which leads the model relies on, shown in Figure 4.13. The aggregate ranking placed aVR highest (mean attribution 0.0026), followed by V6 (0.0022), V1 (0.0020), Lead II (0.0019), and Lead I (0.0018), with the augmented limb leads aVF, aVL and the precordial leads V3, V4, together with Lead III, contributing least.

This matters because it shows the single-case finding was not an accident. The same leads that dominated the individual example - aVR, V6, V1, II, I- are precisely the five that rise to the top when averaged across twenty cases, even if their internal order shifts slightly. The model, in other words, has learned a consistent and repeatable preference for a particular subset of leads when judging atrial fibrillation, rather than attending to different arbitrary regions from one recording to the next. That consistency is itself a modest form of validation. A model keying on noise would not concentrate its attribution so reliably on the same physiologically relevant leads across a population of cases.

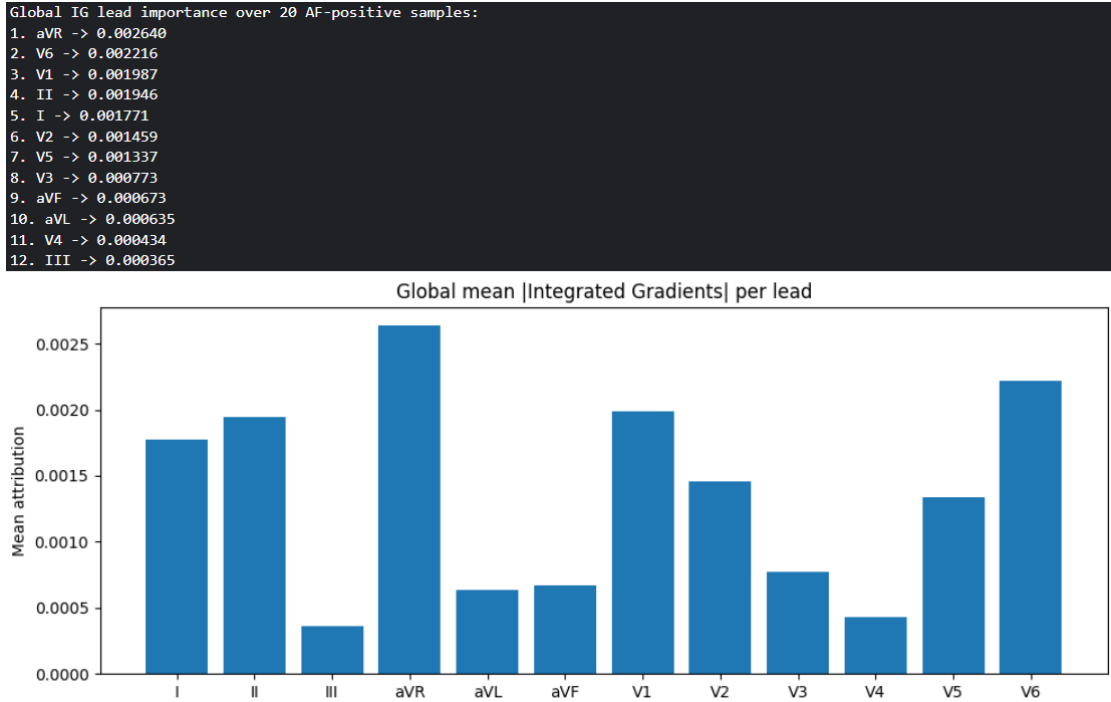


Figure 4.13: Global mean absolute Integrated Gradients attribution per lead, averaged over twenty atrial-fibrillation-positive recordings. The limb lead aVR and the precordial leads V6 and V1 contribute most.

4.7.4 Faithfulness of the Attributions

A consistent ranking across methods shows that they agree, but agreement alone does not establish that the highlighted regions actually drive the prediction. To test this directly, a deletion-insertion analysis was run. In the deletion test, time points are removed from the signal in order of decreasing attributed importance while the atrial-fibrillation probability is tracked. A faithful attribution causes the probability to fall quickly, yielding a low area under the deletion curve. The insertion test is the mirror image—the most important points are added back to an empty baseline first—where a faithful attribution drives the probability up quickly, yielding a high area under the insertion curve. A random-importance ordering serves as a control.

Averaged over thirty atrial-fibrillation-positive recordings (Table 4.5), every attribution method improved on the random control, confirming that the regions they highlight genuinely influence the model rather than merely coinciding with its decision. Occlusion was the most faithful by the deletion measure. Driving the predicted probability down fastest as its top-ranked points were removed (deletion AUC 0.570 against the control’s 0.722). While Integrated Gradients was the strongest on insertion, recovering a confident prediction from the fewest points (0.891 against 0.725). Grad-CAM was the weakest of the four, with an insertion score close to the random baseline—an expected consequence of its being a coarse, upsampled map rather than a point-level attribution. The fused map, which averages the

normalised methods, sat between its components on both measures. Taken with the cross-method agreement reported above, these quantitative results give the attribution evidence a firmer footing than visual inspection alone.

Table 4.5: Faithfulness of each attribution method, measured by deletion and insertion area under the curve and averaged over thirty atrial-fibrillation-positive test recordings. Lower deletion AUC and higher insertion AUC both indicate a more faithful attribution; the random ordering is a control.

Method	Deletion AUC ($\downarrow$)	Insertion AUC ($\uparrow$)
Integrated Gradients	0.712	0.891
Grad-CAM	0.760	0.715
Occlusion	0.570	0.868
Fused	0.721	0.825
Random (control)	0.722	0.725

4.7.5 Natural-Language Explanations

The final step turned these attribution findings into short readable explanations through a template-based generator. The aim was not a polished clinical report but a plain statement of what the model saw and where, so that the attribution output could be read without interpreting a heatmap directly. The generator works from a structured record: the predicted AFIB probability, the top contributing leads, and the most influential time windows are assembled into an input block, and a deterministic template verbalises them into an explanation. For the true-positive case discussed above, that record carried the predicted probability of 0.989, the top leads V1, aVR, and Lead II, and the windows 4.00–4.50 s, 8.50–9.00 s, and 7.50–8.00 s, yielding the following explanation:

The model predicts AF with high confidence (predicted probability = 0.989). The strongest evidence is concentrated in leads V1, aVR, I. The most influential time regions are around 4.00–4.50 s, 8.50–9.00 s. These regions were consistently highlighted by input-gradient and perturbation-based explainability methods.

```

INPUT:
AF probability: 0.9886
Top leads: V1, aVR, II
Top windows: 4.00-4.50s; 8.50-9.00s; 7.50-8.00s
Task: Write a short ECG-AF explanation.

TARGET:
The model predicts AF with high confidence (predicted probability = 0.989). The strongest evidence is concentrated in leads V1, aVR, I. The most influential time regions are around 4.00-4.50s, 8.50-9.00s. These regions were consistently highlighted by input-gradient and perturbation-based explainability methods.

```

Figure 4.14: Natural language explanation example

case index: 525 | true: 1 | prob: 0.999

[TP] $P(AF)=0.999$

Top leads: ['V1', 'I', 'V5']

Top windows: ['9.10-9.60s', '2.83-3.33s', '6.60-7.10s']

Explanation:

The model predicts atrial fibrillation with high confidence ($P(AF) = 0.999$). The strongest lead-level evidence is in V1, I, V5, and the most influential time regions are around 9.10-9.60s, 2.83-3.33s. These regions were consistently highlighted by gradient-based (Integrated Gradients, Grad-CAM) and perturbation-based (occlusion) explainability methods.

Figure 4.15: Another natural language explanation example

This output is faithful to the underlying analysis, which is the main thing to ask of it. And because the generator is template-based, that faithfulness holds by construction, every phrase is produced directly from the computed evidence. The stated probability of 0.989 matches the model’s actual prediction. The named leads correspond to the top-ranked attribution leads, and the cited time windows are the highest-scoring intervals identified by the windowing step. The explanation summarises the attribution rather than inventing detail beyond it.

Two honest qualifications belong here. First, the text describes *where* the model found evidence, not *why* that evidence is clinically diagnostic of atrial fibrillation; phrases like “strongest evidence” refer to attribution magnitude, not to a verified physiological finding, and a reader should not mistake the fluency of the sentence for clinical confirmation. Second, the present generator is deterministic rather than learned - its faithfulness is also its ceiling. Since it can only restate the attribution in fixed phrasing and cannot adapt its wording or generalise. This was a deliberate choice. As template guarantees accuracy where a freely generating model might embellish. To support a learned generator without sacrificing that discipline, each explanation is built as an instruction-style input-target pair, leaving the data ready for parameter-efficient fine-tuning- for instance with Low-Rank Adaptation - as a future extension. Used as intended, as a plain-language window onto the attribution rather than a standalone diagnosis, the current generator does its job: a reader can grasp what drove the prediction without having to interpret a twelve-lead heatmap directly.

4.7.6 Explainability on the External Set

The explainability analysis to this point was carried out entirely on PTB-XL recordings, the same distribution the detector was trained on. A fair question is whether the model’s attribution behaviour transfers to the external set as cleanly as its detection performance did. If the four methods continue to converge on clinically sensible leads when applied to a Chapman recording the model has never seen, the interpretability claim gains the same external footing as the ROC-AUC result in Section 4.6. If instead the attribution scatters or fixes on incidental regions, that would temper how far the explainability finding should be read.

To test this, the full attribution pipeline-Integrated Gradients, Grad-CAM, occlusion, and feature ablation -was applied, with the frozen epoch-8 model and the identical pre-processing, to a correctly detected Chapman atrial-fibrillation case (case 525), which the model flagged with an AFIB probability of 0.999. The picture is broadly consistent with the internal analysis, with one honest difference worth stating plainly. The attribution again concentrates on a small set of leads rather than spreading across all twelve. And the temporal importance again clusters around the individual beats rather than the quiescent stretches between them; the same beat-aligned pattern seen on PTB-XL. The leading leads for this case were V1, Lead I, and V5. V1 remains at or near the top, exactly as on the internal data and exactly where a clinician would look for atrial activity, which is reassuring. The shift is that aVR and V6, prominent in the PTB-XL analysis, give way here to Lead I and V5, while Lead II again features among the temporally important regions.

This shift should not be overstated in either direction. It is not evidence that the model attends to clinically irrelevant leads. V1, I, II, and V5 are all reasonable leads for a rhythm judgement, and the precordial-plus-limb mix is qualitatively the same kind of evidence surfaced internally. But neither is it identical to the PTB-XL ranking, and reporting it as such would be misleading. The most likely reading is that the precise lead ordering is somewhat sensitive to the recording, the population, and the acquisition setup. while the higher-level behaviour -concentrating on a few rhythm-relevant leads and on beat-aligned time windows rather than on artefacts- is the part that transfers. That is the more important property for an interpretability claim, and it is the part that holds.

As on the internal case, the agreement *across* methods is what gives the finding weight. Integrated Gradients, Grad-CAM, occlusion, and feature ablation -built on accumulated gradients, convolutional-layer activations, and input perturbation respectively-point to the same leads and the same beat-aligned timing on this external recording, just as they did internally (Figure 4.16). The fused view across all four methods (Figure 4.17) shows the same QRS-aligned concentration, and the per-lead summary (Figure 4.19) places V1 first across every method. The template generator, run on this case, produced an explanation naming the top leads V1, I, V5 and the most influential windows around 8.10–8.60 s, 9.10–9.60 s, and 7.60–8.10 s, faithful by construction to the computed attribution in the same way as the internal example:

The model predicts atrial fibrillation with high confidence ($P(AF) = 0.999$). The strongest lead-level evidence is in V1, I, V5, and the most influential time regions are around 8.10–8.60 s, 9.10–9.60 s. These regions were consistently highlighted by gradient-based (Integrated Gradients, Grad-CAM) and perturbation-based (occlusion, feature ablation) explainability methods.

One qualification bounds this result firmly, it rests on a single external case. where the internal analysis was strengthened by averaging Integrated Gradients over twenty AFIB-

positive recordings (Section 4.7.3). A single tidy case on Chapman is suggestive, not conclusive. And is open to exactly the ‘it could be the one case that looks neat’ objection raised earlier. Averaging the external attribution over a comparable set of Chapman AFIB-positives would put the external interpretability claim on the same footing as the internal one, and is a natural and inexpensive extension. As it stands, the external case is best read as a consistency check that the attribution behaviour did not collapse off-distribution, rather than as an independent population-level result.

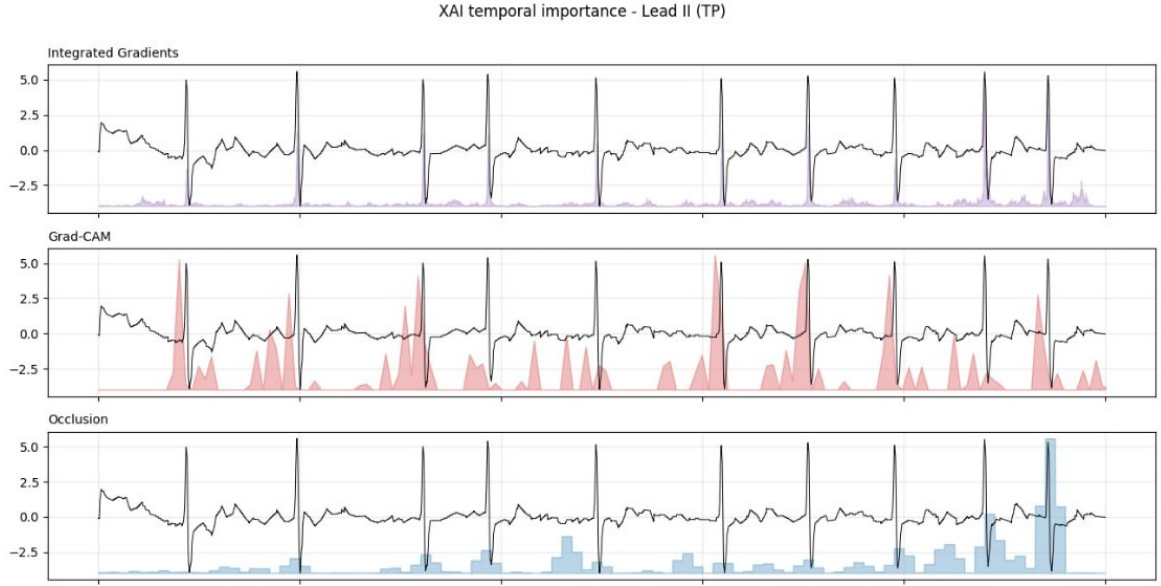


Figure 4.16: XAI temporal importance over Lead II for a correctly detected Chapman atrial-fibrillation case (case 525, AFIB probability 0.999), showing Integrated Gradients, Grad-CAM, and occlusion. Shaded importance concentrates around the individual beats, echoing the QRS-aligned pattern seen on PTB-XL.

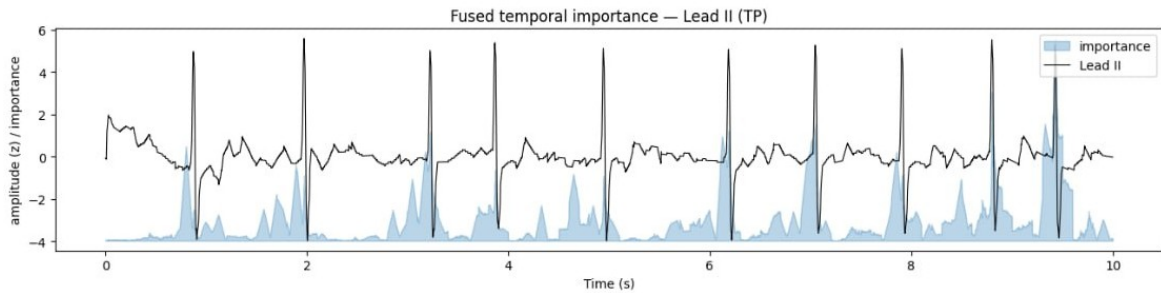


Figure 4.17: Fused temporal importance over Lead II for the same external case, averaging the four attribution methods. The beat-aligned concentration matches the internal true-positive analysis.

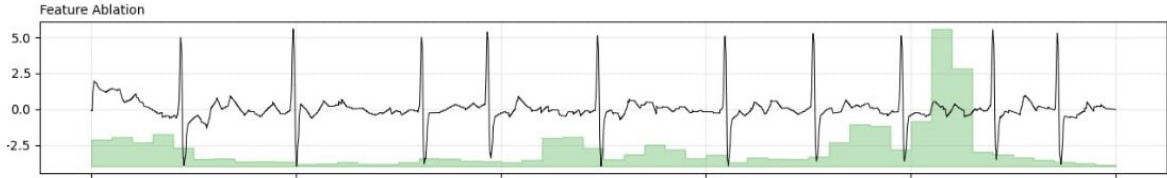


Figure 4.18: Feature ablation temporal importance over Lead II for the same external Chapman case. Importance again concentrates around the individual beats, in agreement with the other three methods.

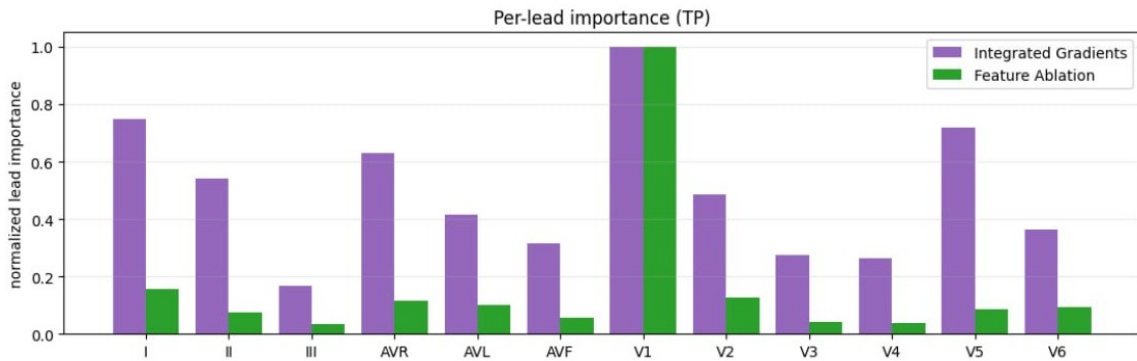


Figure 4.19: Figure 4.16: Per-lead importance for the external Chapman case (case 525), normalised per method, shown for Integrated Gradients and Feature Ablation. V1 ranks first for both methods, with Lead I and V5 also prominent; this ordering is consistent with the occlusion and Grad-CAM results in Figures 4.13–4.15, so V1 leads across all four attribution methods

4.7.7 Clinical Coherence Explanation

The lead ranking in Figure 4.19 is clinically coherent. V1 ranks first, which is the physiologically expected result for an atrial-fibrillation decision: V1 overlies the right atrium under standard 12-lead electrode placement [40] and is the surface lead where fibrillatory waves and the absence of organised P-waves are most visible, so it is the lead clinicians rely on to characterise atrial activity and confirm AF [39, 41]. That both methods shown—and, in Figures 4.16–4.17, occlusion and Grad-CAM as well—place V1 at the top indicates the model is keying on the standard electrophysiological substrate of the arrhythmia rather than on an incidental lead. Lead I and V5 follow: Lead I is a standard limb lead for judging the irregularly irregular ventricular response that characterises AF [41, 42], and V5 is a lateral precordial lead with a tall, clean R-wave in which QRS complexes are crisply timed, making both sensible places to read beat-to-beat rhythm regularity. The set as a whole is a physiologically reasonable mix—V1 for atrial evidence together with limb and lateral precordial leads for ventricular timing—rather than a scatter across electrically redundant or noise-prone leads. Compared with the internal PTB-XL ranking (V1, aVR, II, I, V6), the

external case retains V1 at the top and stays within the same rhythm-relevant family, with aVR and V6 giving way to Lead I and V5; the exact ordering shifts with population and acquisition setup while the higher-level behaviour—concentrating on a few rhythm-relevant leads—transfers, which is the property that matters for an interpretability claim. As elsewhere, this alignment shows the model attends to clinically sensible leads; it is supporting evidence, not proof that the model’s internal reasoning is clinically correct [21].

4.8 Discussion

Taken as a whole, the results support the thesis’s central claim. The detector identifies atrial fibrillation reliably - a test ROC-AUC of 0.979 and 93% recall on a class making up under 7% of the data is a solid result, and it shows that a compact CNN-BiLSTM trained with imbalance-aware sampling can pick out a rare arrhythmia without resorting to a much larger model. Just as importantly, that performance did not collapse when the model was moved to an entirely separate population. On the external Chapman-Shaoxing data the ROC-AUC held at 0.961 and recall stayed high, so the detector’s ability to rank and catch arrhythmia cases transferred rather than being an artefact of the training distribution. The precision-recall balance was deliberately set to favour recall, which suits a screening role, and can be retuned through the decision threshold for settings that weight the two errors differently.

The more distinctive contribution is that the detector does not ask to be taken on trust. The attribution analysis showed, across four independent methods and on both a single case and a population of twenty, that the model’s decisions rest consistently on a small set of clinically sensible leads- V1, aVR, Lead II, Lead I, and V6 recur throughout - and on the regions around individual heartbeats. The agreement among methods built on different principles is what gives that finding its weight, and the template-based generator turns it into something a non-specialist reader can follow. Together, detection and explanation make the case that accuracy and interpretability need not be traded against each other in this setting.

The diagnostic head is best viewed as a secondary function that is trained together with the main encoder, and that gives a useful diagnostic rank (macro-AUC 0.908) and also provides a secondary learning signal that influences the representation that is learned by the main AF detector.

4.8.1 Limitations

It is worth gathering the limitations of this work in one place rather than leaving them scattered. since an honest account of where a model falls short is as useful as the headline result. Three stand out.

First, the figures reported here rest on two retrospective datasets - PTB-XL for development and Chapman-Shaoxing for external testing. The external evaluation is a genuine strength, since it shows the detector transfers to an unseen population rather than merely to a

held-out slice of its own training data. But two datasets still do not span the range of equipment, populations, and clinical contexts a deployed system would meet, and the precision drop observed on Chapman is a plain reminder that transfer is not free. The strong detection performance should therefore be read as evidence across two settings; not as a guarantee of general clinical robustness.

Second, the detector was deliberately tuned to favour recall. Which suits a screening role but leaves its precision lower- roughly a quarter of its positive calls are false alarms at the chosen operating threshold, and a larger fraction on the external set. The operating point reported here is thus a choice rather than a fixed property of the model. A different deployment would demand a different balance, and, as the external results showed, a threshold tuned on one population does not transfer perfectly to another and would benefit from local recalibration.

Third, the explainability analysis establishes where the model attended, not whether its reasoning is clinically correct. The consistency of the attribution across four methods is encouraging, and its alignment with rhythm-relevant leads is suggestive. But neither amounts to confirmation by a cardiologist, and the generated natural-language explanations inherit the same limit. They faithfully summarise the attribution without validating it against expert clinical judgement. None of these undermines the central result, but each bounds how far it should be read, and each points to a concrete line of further work taken up in Chapter 5.

CHAPTER 5:

EVALUATION

A working model is not the end of the story. A piece of engineering that touches health data and produces decisions about a person's heart carries obligations that go well beyond accuracy figures, and this chapter steps back to weigh them. It assesses the societal, health, safety, legal, and cultural dimensions of the work, considers its environmental and sustainability footprint, sets out the ethical principles followed during the research, and closes with the directions the results most naturally point toward.

5.1 Assessment of Issues

5.1.1 Societal Issues

The social promise of a tool such as this is simple. The automated ECG screening could be extended to. A tool used in cardiac evaluation to areas where a trained cardiologist is not readily available. Hand, covering a vast amount of the world. A model to identify likely atrial fibrillation. could assist in triaging patients, determining priority of patient treatment or providing a second read by a specialist. time is scarce. It's indeed a social good worth considering.

Displays that same capacity come with a social risk that must be spoken of in the same breath. A model is Neither more nor less representative than the data it is fed. The screening tool trained on one population may perform poorly in that data as a group, but this can be done quietly without any disruption. profiles, differences in prevalence of conditions, different recording equipment. Deployed incompetent health workers a platform to expand, not to contract, inequalities.If not properly used, such a tool could exacerbate, rather than reduce, health inequities, and could provide a stage for the arrogant but incompetent health worker to speak ill of these inequalities. To deliver less accurate readings to those populations already underserved. The upright stance is This work is a research prototype, not a commercial product. It was trained and tested on a. These high detection rates are only to be understood within the context of a single dataset, and are not intended to be interpreted as evidence. But not necessarily a promise of consistent clinical results when not supervised by trained professionals.

5.1.2 Health and Safety Issues

The central health-and-safety consideration for any diagnostic aid is the asymmetry between its two kinds of error. A false negative- missing a real abnormality - can delay treatment and is generally the more dangerous failure. A false positive - flagging a healthy recording - causes unnecessary worry and follow-up but is rarely harmful in the same way. The design of this system deliberately leaned toward catching the rare, important cases, and the results bear that out. The model recalled roughly 93% of true AFIB cases on the test set, accepting a higher false-alarm rate as the price. For a screening role that is the defensible trade.

That same trade is also the reason the system must remain an assistant rather than an authority. A recall-oriented detector raises a meaningful number of false positives, and acting on a positive flag without expert confirmation could lead to unnecessary follow-up or undue alarm. The safe framing, and the one this thesis adopts, is that the model supports a clinician’s judgement by surfacing candidates and the evidence behind them. And never replaces the clinician’s decision.

5.1.3 Legal and Cultural Issues

On the legal side, the principal concern in medical machine learning is the handling of patient data. This work was built on publicly available, de-identified electrocardiography datasets. In which personal identifiers have already been removed by the data providers. so it did not involve collecting or processing identifiable patient records directly. Any extension of the work to additional data sources would need to respect the licensing terms and de-identification guarantees of those sources, and any eventual clinical use would fall under the medical-device and data-protection regulations of the jurisdiction in which it operated – a bar far higher than that for a research study, and one this work does not claim to meet.

Culturally, the use of automated tools in medicine sits alongside, rather than above, established clinical practice and the trust between patient and physician. A diagnostic aid is most appropriately positioned as informing that relationship, not displacing it, and its outputs should always be open to a clinician’s override.

5.2 Evaluation of Environment and Sustainability

The environmental cost of machine learning is increasingly part of responsible practice, and it was a consideration in the design rather than an afterthought. Two choices kept the footprint modest. The model was deliberately compact - a convolutional encoder with a two-layer recurrent stage rather than a very large network- so that training completed in a manageable time. With each epoch taking on the order of a minute or so on a single GPU and the whole run, including early stopping, finishing in well under an hour. The explanation component reinforced the same principle. Rather than fine-tuning a large language model

to generate explanations, the work uses a lightweight template-based generator that runs at negligible cost. While also leaving the attribution data structured so that an efficient adapter such as Low-Rank Adaptation could be trained later if a learned generator were wanted - deferring rather than incurring that expense.

Beyond the training cost, the broader sustainability case for this kind of tool is indirect but real. Software models impose no per-use material cost once trained, and an automated first read could in principle reduce redundant testing and make better use of scarce specialist time. These are modest benefits and should not be overstated, but they point in the right direction.

5.3 Ethical Issues

The research followed the ordinary principles of responsible conduct throughout. All datasets used were public and de-identified, accessed and used in accordance with their terms. So the question of patient consent was handled upstream by the data providers rather than left to this project. No attempt was made to re-identify individuals, and no identifiable health information was collected by the authors.

On academic integrity, all sources, prior work, and external tools and libraries have been cited where used. And the results reported in Chapter 4 are the genuine outputs of the experiments rather than selected or altered figures. The less flattering aspects of the model- for instance the lower precision that comes with its recall-oriented operating point; are reported plainly rather than omitted. Reporting a model's limitations honestly is itself an ethical obligation in work that could one day inform a medical decision, and this thesis has tried to meet it.

5.4 Future Works

The results in this thesis point to several clear next steps, and it is more useful to name the ones the work directly motivates than to list every conceivable extension.

The most immediate concerns the operating point. The detector was tuned to favour recall. which suits a screening role but leaves its precision lower than it could be. A natural refinement is to calibrate the decision threshold deliberately- choosing the point on the precision-recall curve that best fits the intended use- rather than relying on the default. And to report performance across a range of thresholds so that a deployer can pick the balance their setting demands.

A second direction concerns generalisation. Everything reported here rests on a PTB-XL and Chapman-Shaoxing datasets. So the natural next step is to validate the detector on additional, more diverse ECG datasets to test how well its performance transfers across different equipment and patient populations. A model that holds up across sources would be

far more trustworthy than one tuned to a single collection.

Third, the explainability findings invite proper clinical validation. The attribution analysis showed that the model’s predictions consistently rest on rhythm-relevant leads. Which is encouraging, but agreement among attribution methods is not the same as confirmation by a cardiologist. Having clinicians review whether the highlighted leads and time windows truly correspond to the features they would use would turn a suggestive result into a validated one. And would also test whether the generated natural-language explanations are genuinely useful at the bedside or merely plausible-sounding.

Finally, there is room to broaden the scope of the task itself- extending from atrial fibrillation toward other arrhythmias and the broader set of rhythm and morphology findings that large ECG datasets provide. Each of these would build on the imbalance-aware and explainable foundation established here rather than replacing it.

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**Multi-Task Deep Learning for Atrial Fibrillation Detection and
Diagnostic Classification of 12-Lead ECG with Explainable Attribution**

Sadman Samin Rahul

Md. Shahriar Hasan

Jun

2026
